



When adaptation becomes maladaptation: drought-groundwater feedbacks and resilience in agricultural socio-ecological systems

Hıdır Serkendiz¹ 

Received: 22 April 2026 / Accepted: 1 September 2026
© The Author(s) 2026

Abstract

This study investigates how increasing drought pressure, groundwater decline, and rapid land use/land cover (LULC) change reshape the resilience and feedback dynamics of an agricultural socio-ecological system. An integrated framework combining hydroclimatic, vegetation, water, soil moisture, groundwater, and LULC indicators was used to assess long-term changes and critical system transitions. Despite intensifying drought and persistent groundwater decline, vegetation greenness increased, indicating that agricultural production has been sustained beyond natural hydroclimatic limits through human intervention, particularly irrigation. Critical thresholds were identified around 2004 for drought conditions, 2006 for groundwater levels, and 2008 for surface water and vegetation dynamics, with substantial LULC changes during 2006–2012 further indicating an intensification of system transformation. Together, these changes suggest that groundwater-supported agricultural adaptation has increasingly generated maladaptive feedbacks, shifting the system from its historical state toward a fragile alternative regime. The system has not yet collapsed but appears to be operating close to its resilience limits. These findings highlight how adaptation strategies that sustain agricultural production in the short term can undermine long-term socio-ecological resilience when they depend on continued groundwater depletion. Redirecting water and agricultural management toward reducing groundwater dependence is therefore critical to avoiding further degradation and potentially irreversible system change.

Keywords Drought · Agriculture resilience · Groundwater · Socio-ecological system · Resilience · Adaptive capacity · LULC

Introduction

Drought is widely regarded as one of the major natural hazards, exerting multidimensional and profound impacts on both natural environments and human systems (Mishra and Singh 2010). According to analyses based on Emergency Events Database (EM-DAT) data conducted by Donatti et

al. (2024), drought affected approximately 1.4 billion people worldwide between 2000 and 2020, making it the climate-related hazard that impacted the largest number of people during this period. Considering the influence of anthropogenic climate change, both the frequency and severity of drought are projected to increase in the coming decades. Indeed, the IPCC (2023) emphasizes that, driven by changes in the climate system, the frequency, duration, and spatial extent of drought are highly likely to increase across many regions. Moreover, drought is increasingly interacting with heatwaves and wildfires to generate compound and cascading hazards (Yu et al. 2022), thereby further intensifying its ecological, economic, and societal impacts.

Agriculture is among the sectors in which the impacts of drought are most pronounced. This is primarily because soil moisture is one of the first components of the hydrological system to be affected by drought, and because both the timing and amount of precipitation during the growing season directly determine agricultural productivity (Wilhite

Handled by Mohammad Valipour, Metropolitan State University of Denver, United States of America.

This study was conceived and the original manuscript was prepared and submitted while the author was working as an independent researcher in Çanakkale, 17020, Türkiye.

✉ Hıdır Serkendiz
hserkendiz@gmail.com; H.Serkendiz@exeter.ac.uk

¹ Department of Geography and Global Systems Institute, University of Exeter, Laver Building, North Park Road, Exeter, Devon EX4 4QE, UK

2000). In this context, regions whose economic structures or livelihoods are largely dependent on agriculture emerge as the most vulnerable to drought (Koley and Jeganathan 2022; Ortega-Gaucin et al. 2018). Therefore, understanding the impacts of drought on agricultural systems is of critical importance for assessing the adaptive capacity, resilience mechanisms, and long-term sustainability of agro-ecosystems.

In this context, the socio-ecological system (SES) approach provides a robust analytical framework for comprehensively assessing the impacts of drought on agricultural systems. Gallopin (2006) argue that, from a sustainable development research perspective, the natural unit of analysis is the socio-ecological system, which integrates interacting human and ecological subsystems. The impacts of drought, considered a fundamental perturbation for agricultural systems, can be meaningfully captured not only through biophysical processes but also when examined in conjunction with social, economic, and governance dynamics (Hossain et al. 2016). Indeed, the SES approach conceptualizes complex social and ecological components as an interconnected system, enabling the analysis of causal relationships, feedback mechanisms, non-linear responses, and spatio-temporal dynamics (Roy et al. 2025). Through these attributes, the SES framework allows the impacts of drought on agriculture to be evaluated not merely in terms of production losses, but also in relation to the system's adaptive capacity and resilience characteristics. The resilience perspective extends beyond the ability of systems to absorb shocks to encompass their potential for renewal, reorganization, and transformation (Folke 2006), which is particularly critical for understanding how agricultural socio-ecological systems may respond to future stresses under climate change. Therefore, an SES-based assessment that jointly considers social and ecological processes operating across multiple spatial and temporal scales offers an appropriate and integrative approach for more comprehensively and explicitly elucidating the underlying dynamics of the drought–agriculture interaction (Li et al. 2020; Ostrom 2009).

The explanatory power of the SES approach in the context of drought and agricultural systems has been demonstrated through applied studies conducted across diverse geographical settings and thematic foci. Examining the long-term sustainability of irrigation water as a common-pool resource in a semi-arid environment, Cox (2014) conceptualized acequia irrigation systems as socio-ecological systems and showed that alignment between the physical characteristics of water and social norms, trust, reciprocity, and multi-level governance structures enables collective action. Similarly, Zhu and Wang (2024) demonstrated that, under arid basin conditions, agricultural cooperatives can

reduce ecological pressures by transforming water use and production practices through collective decision-making and institutional mechanisms.

Studies addressing transformations in agricultural land use from an SES perspective further indicate that feedback relationships among production intensification, income gains, and resource use can generate both positive and negative effects on SES (Aranzabal et al. 2008; Chen et al. 2024). In this regard, processes of land-use intensification, agricultural abandonment, and restructuring are emphasized as being driven not only by biophysical constraints but also by social, economic, and institutional forces (Antoni et al. 2019). Overall, this body of literature demonstrates that SES-based analyses, which explicitly account for non-linear interactions, cross-scale linkages, and feedback loops, provide a powerful framework for understanding the vulnerability, resilience, and adaptive dynamics of agricultural systems (Domptail et al. 2013; Oliveira et al. 2022; Young et al. 2006).

Despite these advances, an important gap remains in understanding how drought adaptation itself may reshape the long-term resilience of groundwater-dependent agricultural socio-ecological systems. Previous studies have provided valuable insights into drought impacts, groundwater depletion, agricultural land-use change, and socio-ecological resilience (van Vliet et al. 2015; Rani et al. 2023; Gullacher et al. 2023; Serkendiz et al. 2023); however, these processes are often examined separately or primarily in terms of vulnerability and adaptation outcomes. Consequently, limited attention has been given to the temporal feedbacks through which irrigation-based responses to drought may initially sustain agricultural production while simultaneously increasing groundwater pressure and generating maladaptive dynamics over time (Serkendiz et al. 2025). Moreover, the potential for such coupled drought-groundwater-agriculture interactions to drive threshold behavior and shifts in system resilience remains insufficiently understood. Addressing this gap, the present study integrates long-term hydroclimatic, groundwater, vegetation, and land-use dynamics within a socio-ecological systems framework to identify feedback mechanisms and critical transitions. Its novelty lies in explicitly tracing how an adaptation strategy that buffers agricultural production against drought in the short term can progressively undermine the resource base on which that adaptation depends, thereby transforming adaptation into maladaptation and potentially pushing the system toward an alternative resilience state.

This study is conducted in the Konya Closed Basin (KCB), located in the central Anatolian Peninsula, Türkiye, characterized by semi-arid climatic conditions and widely regarded as a natural laboratory for drought research. As clearly demonstrated by previous studies, the KCB has been

exposed to recurrent and severe drought episodes throughout its historical record, posing a significant climate-related hazard to agricultural production, which constitutes the basin's primary livelihood and has earned the region the designation of Türkiye's "grain basket" (Dogan et al. 2012; Kartal and Nones 2024; Serkendiz et al. 2024, 2025).

Hydrologically, the KCB is a closed basin in which surface water resources for agricultural irrigation are extremely limited, rendering groundwater the principal source of water supply. Consequently, increasing drought conditions adversely affect groundwater recharge on the one hand, while declining soil moisture on the other hand intensifies irrigation demand, thereby exacerbating the imbalance between water supply and demand. Indeed, previous studies have clearly shown that the widespread expansion of irrigated agriculture in the basin, coupled with severe droughts in recent decades, has led to pronounced and persistent declines in groundwater levels (Gedik and Sarış 2026; Kartal and Nones 2024). Despite these trends, the continued cultivation of water-intensive crops such as sugar beet and maize under conditions of declining groundwater availability indicates the persistence of maladaptive agricultural decisions in the basin (Serkendiz et al. 2025). This situation suggests that the groundwater-dependent irrigated agricultural system is increasingly straining the socio-ecological resilience limits of the basin and may give rise to potential threshold behaviors. Within this context, the present study adopts a SES framework to examine drought in the KCB, analyzing the feedback loops, resilience state, and adaptive dynamics of the groundwater-dependent agricultural system, and discussing long-term sustainability in the basin.

Against this background, the present study examines the long-term dynamics of the groundwater-dependent agricultural socio-ecological system of the KCB under increasing drought pressure, with particular emphasis on feedback mechanisms, resilience, and maladaptation. Drought is conceptualized as a primary external perturbation, while changes in groundwater, soil moisture, vegetation water status, vegetation condition, and agricultural land use are evaluated as interconnected responses within the system. Long-term remote sensing and reanalysis time series, trend and regime-shift analyses, lagged relationships, and LULC change detection are integrated to trace the temporal evolution of these interactions.

Specifically, the study aims to: (1) characterize the long-term evolution of drought and associated hydro-ecological conditions in the KCB; (2) identify trends, lagged responses, and potential critical transitions in groundwater, soil moisture, vegetation water status, and vegetation condition under increasing drought pressure; (3) determine how irrigated and rainfed agricultural land-use patterns have changed alongside these biophysical dynamics; and (4) evaluate

whether groundwater-dependent irrigation has shifted from an adaptive response that sustains agricultural production toward a maladaptive feedback that erodes long-term socio-ecological resilience. By linking these objectives within a common SES framework, the study seeks to clarify when and how short-term adaptation to drought may contribute to longer-term systemic vulnerability and potential regime change.

Data and methodology

Study area

The KCB is located in the central part of the Anatolian Peninsula and constitutes Türkiye's largest hydrologically closed basin (Fig. 1a). Covering an area of approximately 50,000 km², the basin is bordered by the provinces of Ankara and Kırşehir to the north, Aksaray and Niğde to the east, Karaman and Antalya to the south, and Isparta and Afyonkarahisar to the west (Fig. 1b). From a physical geography perspective, the KCB is characterized by a relatively simple morphology consisting of extensive plateau surfaces, closed depressions, and low-gradient plains. Elevations within the basin generally range between 900 and 1,100 m, while the surrounding mountainous areas reach altitudes of up to 3,500 m. The low-gradient topography facilitates the spatial expansion of agricultural activities; however, the extremely limited surface water resources and the absence of outflow beyond the basin boundaries have rendered the hydrological system largely dependent on groundwater.

Climatically, the KCB exhibits semi-arid characteristics, where low and irregular precipitation patterns combined with high evaporation rates exert persistent pressure on water resources (Serkendiz et al. 2024). The basin has an annual mean temperature of approximately 11.5 °C and an average annual precipitation of around 300 mm. According to the Köppen-Geiger climate classification, the region is generally classified as having a warm-summer Mediterranean climate (Csb) (Taşoğlu et al. 2024). From a human geography perspective, the KCB is a basin in which agriculture represents the dominant land-use type, and rural economic activities are largely dependent on irrigated farming. CORINE land-use data indicate that agricultural areas occupy a substantial proportion of the basin (Fig. 1c). In particular, the widespread cultivation of water-intensive crops such as sugar beet, maize, and cereals has increased groundwater abstraction, playing a decisive role in shaping the basin's socio-ecological balance (Serkendiz et al. 2025).

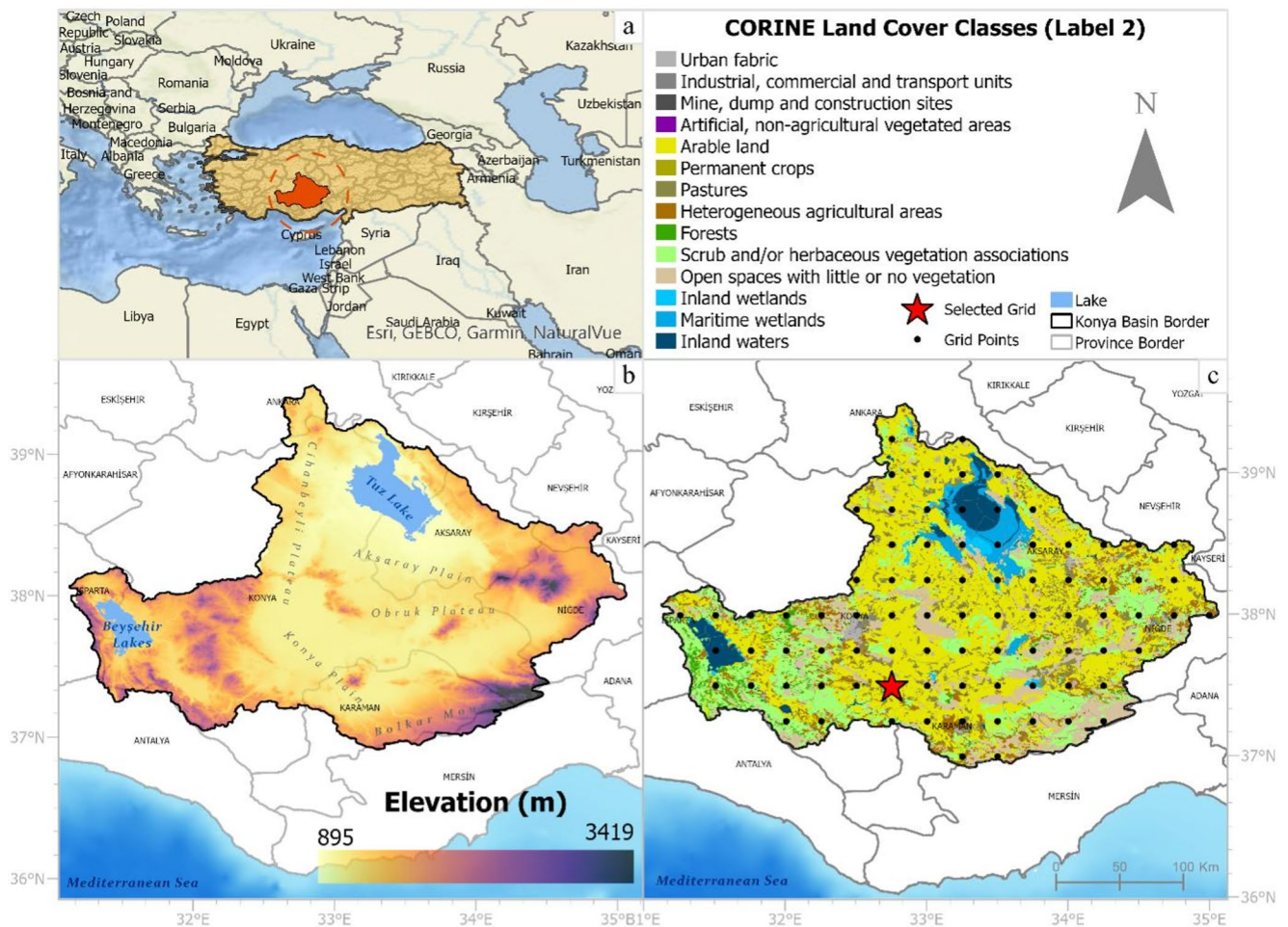


Fig. 1 Map showing the geographical location of the Konya Closed Basin (a), the topographic map illustrating the general morphological characteristics of the basin (b), and the map depicting the main land-use classes based on CORINE land cover data (c)

Data

Table 1 summarizes the main characteristics and sources of the datasets used in this study. The analyses are based on monthly mean air temperature ($^{\circ}\text{C}$), monthly total precipitation (mm), NDVI, NDWI, soil moisture (m^3/m^3), and groundwater (cm) derived from 83 grid points with an approximate spatial resolution of 28 km, representing the KCB. The datasets were downloaded and pre-processed using custom-developed scripts on the Google Earth Engine (GEE) platform, in accordance with the temporal coverage and spatial resolution specified in Table 1. CORINE LULC data employed for the analysis of land-use and land-cover changes were obtained from the official database of the Copernicus Land Monitoring Service.

Monthly groundwater data derived from NASA GRACE Mascon for the 2002–2024 period contained 49 missing monthly observations in each grid cell. These included intermittent gaps within the GRACE record as well as an extended discontinuity associated with the transition

between the GRACE and GRACE Follow-On missions, consistent with known limitations in GRACE temporal coverage (Humphrey et al. 2023). To obtain a temporally continuous dataset, missing groundwater values were estimated using a two-stage gap-filling approach implemented in the R statistical environment. First, a complete Year-Month framework was constructed for each grid cell, and temporal gaps were filled on a grid-by-grid basis using Kalman smoothing as implemented in the *imputeTS* package. This state-space-based approach estimates missing observations from the temporal structure of the available series and was used to maintain temporal continuity and the dominant patterns of groundwater variability. Subsequently, monthly spatial smoothing was applied to reduce residual spatial noise and improve spatial coherence among neighboring grid cells.

The resulting groundwater series should nevertheless be interpreted with consideration of uncertainties associated with both GRACE-derived groundwater estimates and the gap-filling procedure. GRACE observes changes in total

Table 1 Data sources and key characteristics of the datasets used in the study

Variable	Description	Temporal coverage	Spatial resolution	Data format	Data source
Monthly mean air temperature	2 m air temperature (°C)	1985–2025	0.25°	ASCII	ECMWF ERA5-Land
Monthly total precipitation	Total accumulated precipitation (mm)				
Soil moisture	Volumetric soil water content (layer 3, m ³ /m ³)				
NDVI	Derived from red and near-infrared bands	1985–2025	0.25°	ASCII	Landsat 5, 7, and 8
NDWI	Derived from green and near-infrared bands				
Groundwater	Liquid water equivalent (LWE) thickness (cm)	2002–2024	~111 km	ASCII	NASA GRACE Mascon
LULC	CORINE LULC inventory (44 thematic classes)	1990, 2000, 2006, 2012 and 2018	100 m	Raster	Copernicus Land Monitoring Service

terrestrial water storage rather than groundwater directly; groundwater variations are therefore derived by accounting for other hydrological storage components, introducing uncertainties from both GRACE observations and the ancillary data used in this separation (Scanlon et al. 2012; Humphrey et al. 2023). Its relatively coarse spatial resolution further limits the interpretation of highly localized groundwater variability. In addition, Kalman smoothing provides model-based estimates rather than direct observations for missing months and may attenuate short-term or extreme variations, particularly across prolonged gaps; subsequent spatial smoothing may further reduce localized variability. Accordingly, the completed dataset was used primarily to characterize broad and persistent spatio-temporal patterns and long-term groundwater dynamics, and the magnitude and timing of individual groundwater anomalies and trends were interpreted with appropriate caution.

Methodology

The methodological framework of the study integrates multiple data sources and analytical procedures to evaluate drought-driven changes and their implications for the

agricultural socio-ecological system. The overall workflow, from the derivation of biophysical and hydroclimatic indicators to trend, change-point, relationship, and LULC analyses and their integration within the SES framework, is summarized in Fig. 2; the individual analytical procedures are described in detail in the following subsections.

Drought hazard

Meteorological drought conditions were assessed using the SPEI, developed by Vicente-Serrano et al. (2010), which is based on the climatic water balance and explicitly incorporates the effects of temperature-driven evaporative demand. By integrating precipitation and potential evapotranspiration, SPEI provides a more realistic representation of drought processes, particularly under conditions of increasing air temperatures. In this context, its inclusion of the temperature component enables the index to more effectively capture the influence of changes in surface evaporative demand associated with global warming on drought responses (Liu et al. 2021).

The first step in SPEI computation involves calculating the monthly climatic water balance (D), expressed as the difference between precipitation and potential evapotranspiration (Serkendiz et al. 2024):

$$D_i = P_i - PE_i \quad (1)$$

where D_i denotes the water balance for month i , P_i represents monthly total precipitation, and PE_i is the monthly potential evapotranspiration.

Potential evapotranspiration was estimated using the Thornthwaite method, which calculates PE as a function of mean monthly air temperature and latitude-dependent day length:

$$PE = 16 \cdot d_l \left(\frac{10T}{I} \right)^b \quad (2)$$

where T is the mean monthly air temperature (°C), d_l is a correction factor accounting for day length, and I is the annual heat index computed as:

$$I = \sum_{j=1}^{12} \left(\frac{T_j}{5} \right)^{1.541} \quad (3)$$

The empirical exponent b is derived from the heat index using the following polynomial function:

$$b = 6.75 \cdot 10^{-7} I^3 - 7.71 \cdot 10^{-5} I^2 + 1.79 \cdot 10^{-2} I + 0.49 \quad (4)$$

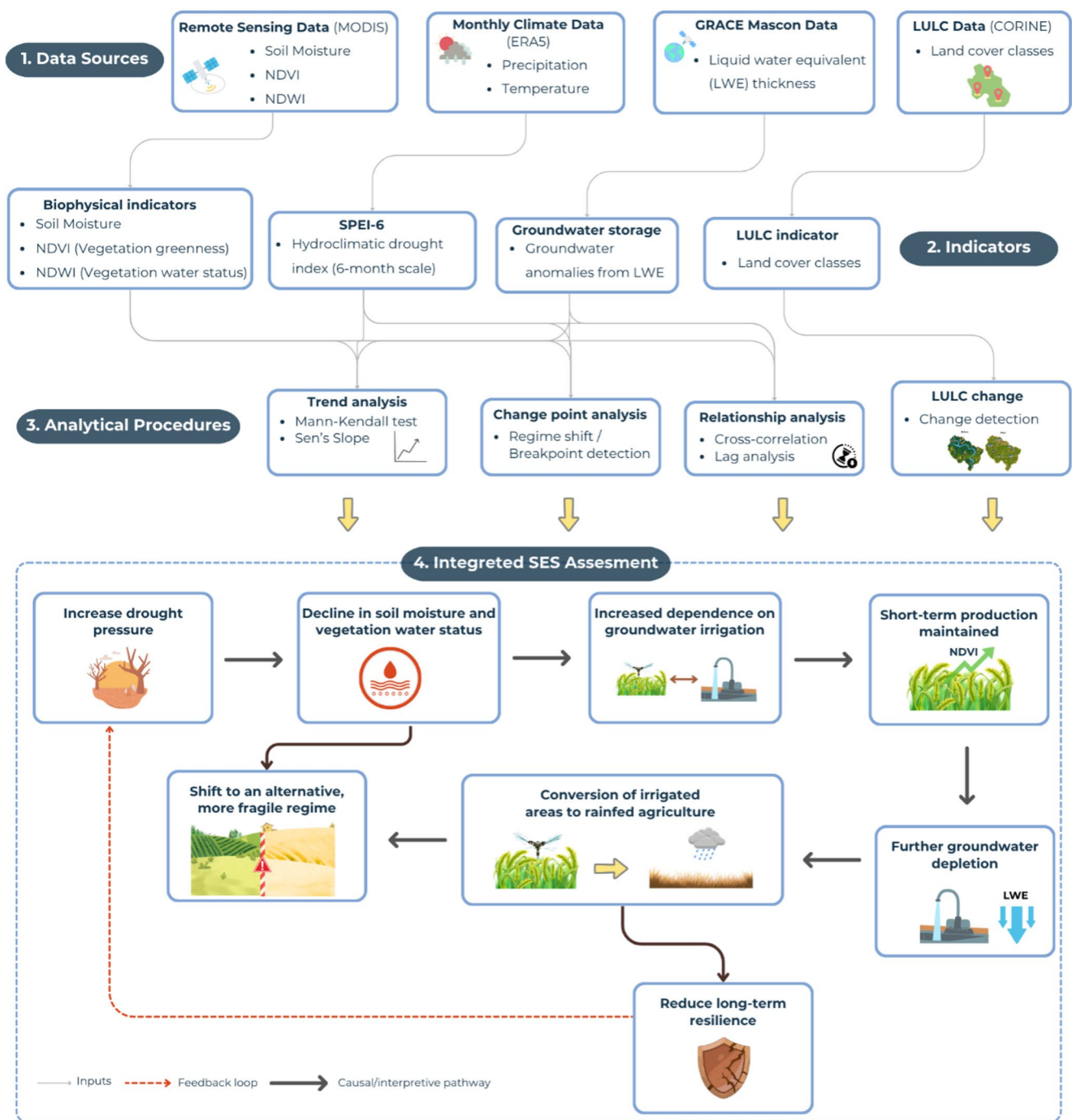


Fig. 2 Methodological workflow integrating data sources, indicators, analytical procedures, and socio-ecological system assessment of drought-groundwater-agriculture feedback, maladaptation, and resilience dynamics in the Konya Closed Basin

After calculating the monthly water balance series, the accumulated D values were aggregated over the selected temporal scale. These aggregated values were then fitted to a log-logistic probability distribution, which has been shown to effectively describe the statistical behavior of climatic water balance series:

$$F(x) = \left[1 + \left(\frac{\alpha}{x - \gamma} \right)^\beta \right]^{-1} \quad (5)$$

where α , β and γ , represent the scale, shape, and location parameters of the distribution, respectively. These parameters were estimated using the L-moment method, which is

robust for hydrometeorological applications (Vicente-Serrano et al. 2010).

The cumulative probability obtained from the fitted distribution was subsequently transformed into standardized units using the inverse normal distribution, yielding the SPEI values:

$$SPEI = W - \frac{C_0 + C_1W + C_2W^2}{1 + d_1W + d_2W^2 + d_3W^3} \quad (6)$$

where W is defined as:

$$W = \begin{cases} \sqrt{-2 \ln(P)} & P \leq 0.5 \\ \sqrt{-2 \ln(1-P)} & P > 0.5 \end{cases} \quad (7)$$

and P denotes the cumulative probability. The constants C_0 , C_1 , C_2 , d_1 , d_2 and d_3 are fixed coefficients used in the approximation of the standard normal distribution.

In this study, SPEI was computed at a 6-month accumulation timescale to represent seasonal to medium-term moisture anomalies that are particularly relevant to agricultural and vegetation responses in the study area. The selection of SPEI-6 was guided by the temporal scale of the processes examined rather than by assuming a universally optimal SPEI accumulation period. Shorter accumulation periods, such as SPEI-3, primarily capture relatively rapid and short-lived moisture anomalies and may therefore be more sensitive to intra-seasonal variability, whereas longer accumulation periods, such as SPEI-12 or SPEI-24, integrate moisture conditions over annual to multi-annual periods and are more appropriate for identifying persistent hydrological drought and longer-term water-resource anomalies. In contrast, the 6-month timescale provides an intermediate window that integrates moisture deficits and atmospheric evaporative demand over a sufficiently sustained period while retaining sensitivity to seasonal drought development relevant to crop growth, vegetation condition, soil moisture, and irrigation demand. This is particularly important because vegetation responses to drought vary across accumulation timescales, with short- to intermediate-term SPEI often showing strong relationships with vegetation activity depending on ecosystem characteristics (Vicente-Serrano et al. 2013). Similarly, agricultural impacts depend not only on drought magnitude but also on its duration and timing within the cropping season, highlighting the importance of assessing moisture deficits at timescales that capture accumulated seasonal stress (Santini et al. 2022). Accordingly, SPEI-6 was selected as a compromise between short-term climatic variability and longer-term hydrological persistence, consistent with the study's objective of tracing drought-related pressures and associated responses within the agricultural socio-ecological system (Vicente-Serrano et al. 2010).

Mann-Kendall trend test

Long-term monotonic trends in hydroclimatic and ecohydrological variables were examined using the MK trend test, a widely applied non-parametric method in environmental and climate-related time series analyses (Hirsch et al. 1991; Kendall 1975; Mann 1945). The MK test is particularly suitable for this study, as it does not require the data to follow a normal distribution and is robust against outliers and missing values, which are common characteristics of environmental datasets.

The MK test was applied to monthly time series of multiple indicators representing different components of the socio-ecological system, including SPEI, NDVI, NDWI, soil moisture, and groundwater storage. Applying the test to monthly data allows for the detection of gradual changes and persistent trends in both climatic forcing and ecosystem and water-resource responses.

The null hypothesis (H_0), assumes that the analyzed time series exhibits no monotonic trend, while the alternative hypothesis (H_1), indicates the presence of a statistically significant increasing or decreasing trend. The test statistic S , is computed based on the relative ranking of all possible data pairs within the time series:

$$S = \sum_{i=1}^{N-1} \sum_{j=i+1}^n \text{sgl}(X_j - X_i) \quad (8)$$

where X_i and X_j denote the values of the variable at times i and j respectively, and the sgl (signal) function is defined as:

$$\text{sgl} = \begin{cases} 1 & x > 0 \\ 0 & x = 0 \\ -1 & x < 0 \end{cases} \quad (9)$$

For time series containing tied values, the variance of the test statistic is calculated as:

$$\text{Var}(S) = \frac{N(N-1)(2N+5) - \sum_{k=1}^m t_k(t_k-1)(2t_k+5)}{18} \quad (10)$$

where N , is the length of the time series, m represents the number of tied groups, and t_k is the number of observations in the k -th tied group.

The standardized test statistic Z is then derived as:

$$Z = \begin{cases} \frac{S-1}{\sqrt{\text{Var}(S)}} & S > 0 \\ 0 & S = 0 \\ \frac{S+1}{\sqrt{\text{Var}(S)}} & S < 0 \end{cases} \quad (11)$$

Trends were considered statistically significant at the 95% confidence level ($\alpha=0.05$). Positive Z values indicate increasing trends, while negative values denote decreasing trends in the analyzed variables.

Following trend analysis, a cross-correlation analysis was conducted between monthly SPEI and NDVI, NDWI, soil moisture, and groundwater storage time series to examine the temporal relationships between meteorological drought and ecohydrological responses. This analysis was employed to quantify both the strength and direction of associations and to identify potential time-lagged responses of environmental variables to drought conditions. Cross-correlation coefficients were calculated for a range of positive time lags, where SPEI was treated as the leading variable and the other indicators as lagged responses. Lag values were expressed in months, representing the delay between the occurrence of drought conditions and the corresponding response in vegetation dynamics, surface moisture, and subsurface water storage. Positive correlation values indicate that increases (or decreases) in SPEI are followed by similar changes in the associated variable after a given lag, whereas negative values reflect inverse responses. The analysis was performed over a lag window of up to 24 months to capture both short-term and delayed system responses. This approach allows

for the identification of differential response times among indicators, reflecting the varying degrees of sensitivity and inertia within the socio-ecological system (Pasho et al. 2012). The resulting cross-correlation functions provide a basis for interpreting the temporal propagation of drought impacts from atmospheric forcing to ecological and hydrological components.

Because the MK test assumes that observations are independent and randomly ordered, serial autocorrelation in environmental time series may affect the statistical significance of detected trends. In particular, positive autocorrelation can increase the probability of identifying a statistically significant trend when no actual trend exists (Hamed and Rao 1998). At the same time, conventional pre-whitening may remove part of an existing trend together with serial dependence, potentially reducing the power of the MK test (Yue and Wang 2002). To evaluate the robustness of the trend results to this issue, an additional sensitivity analysis was therefore performed using trend-free pre-whitening (TFPW). For each grid-cell time series, the original trend was first estimated using Sen's slope and removed from the series, after which lag-1 autocorrelation was evaluated on the detrended series. Where significant serial dependence was detected, the detrended series was pre-whitened, and the estimated trend was subsequently restored before repeating the MK test. The original and TFPW-adjusted results were then compared in terms of trend direction, magnitude, and statistical significance to assess whether serial autocorrelation materially influenced the identified trend patterns.

Table 2 Reclassification of CORINE land use/land cover classes used in the change detection analysis

CORINE code	CORINE level-3 class	New class code	New class name
211	Non-irrigated arable land	1	Rainfed agriculture
212	Permanently irrigated land	2	Irrigated agriculture
213	Rice fields	3	Permanent crops
221	Vineyards		
222	Fruit trees and berry plantations		
223	Olive groves	4	Mixed systems
231	Pastures		
241	Annual crops associated with permanent crops		
242	Complex cultivation patterns		
243	Land principally occupied by agriculture with significant areas of natural vegetation	5	Natural vegetation
244	Agro-forestry areas		
311–324	Forests and semi-natural vegetation	6	Bare / degraded land
331–334	Bare and sparsely vegetated areas		
All remaining classes (urban, wetlands, water bodies, glaciers, NoData)	-	0	Other

Land use/land cover change detection analysis

LULC changes were analyzed using CORINE CLC datasets for the years 1990, 2000, 2006, 2012, and 2018. All spatial analyses were conducted in ArcGIS Pro, employing the Change Detection tool to quantify inter-period transitions in land use patterns. Given that the primary objective of this analysis was to identify changes between irrigated and rainfed agricultural systems, the original CORINE land cover classes were first reclassified into a reduced number of functionally meaningful categories. The reclassification process was carried out using the *Reclassify tool*, aggregating the original CORINE Level-3 classes into six generalized land use classes representing dominant land-use systems. All non-relevant categories, including urban areas, wetlands, water bodies, and miscellaneous land uses, were grouped under an “Other” category to minimize noise and enhance interpretability (Table 2).

The resulting reclassified land use map consists of the following six classes: *Rainfed agriculture*, *irrigated agriculture*, *permanent crops*, *mixed agricultural systems*, *natural vegetation*, and *bare/degraded land* (Table 2). This

classification scheme allows for a focused assessment of agricultural land dynamics while maintaining ecological context through the inclusion of natural and degraded land classes. Subsequently, change detection analysis was performed between consecutive time periods using the reclassified datasets. Land use transitions were quantified for each period, and area-based change matrices were generated to determine the magnitude and direction of land use conversions among the six classes. Inter-period land use change amounts were calculated in terms of areal extent, enabling a comparative evaluation of temporal shifts in irrigated and rainfed agricultural lands and their interactions with other land use types.

All statistical analyses and data processing were performed using the R programming environment, where custom scripts were developed to compute drought indices, trend analyses, cross-correlation functions, and temporal change metrics. Spatial processing, land use reclassification, change detection analyses, and cartographic visualizations were conducted using ArcGIS Pro.

Results

Spatiotemporal evolution of drought conditions

Figure 3 illustrates the temporal variability of SPEI, soil moisture, NDVI, and NDWI across the KCB during the 1985–2025 period. The results indicate a pronounced intensification of meteorological drought conditions across the basin, particularly after 2005. Following this period, SPEI values exhibit more frequent and more severe negative anomalies. Focusing on recent years, 2024 and 2025 are identified as the two driest years within the approximately 40-year analysis period. The intensification of drought is not limited to the temporal dimension but is also evident spatially, as clearly demonstrated by the SPEI spatial distribution maps for the 1985–2025 period presented in Fig. S1. In particular, the spatial extent of drought across the basin has progressively expanded since 2005, reaching extreme levels in certain years. In this context, the years 2007, 2008, 2010, 2013, 2014, 2016, 2018, 2020, 2021, 2024, and 2025 are identified as years during which drought conditions stand out in terms of both severity and spatial extent.

The findings related to drought severity and duration presented in Fig. 4 further elucidate the spatial patterns of this trend. Accordingly, both the severity and duration of drought reach maximum levels particularly in the

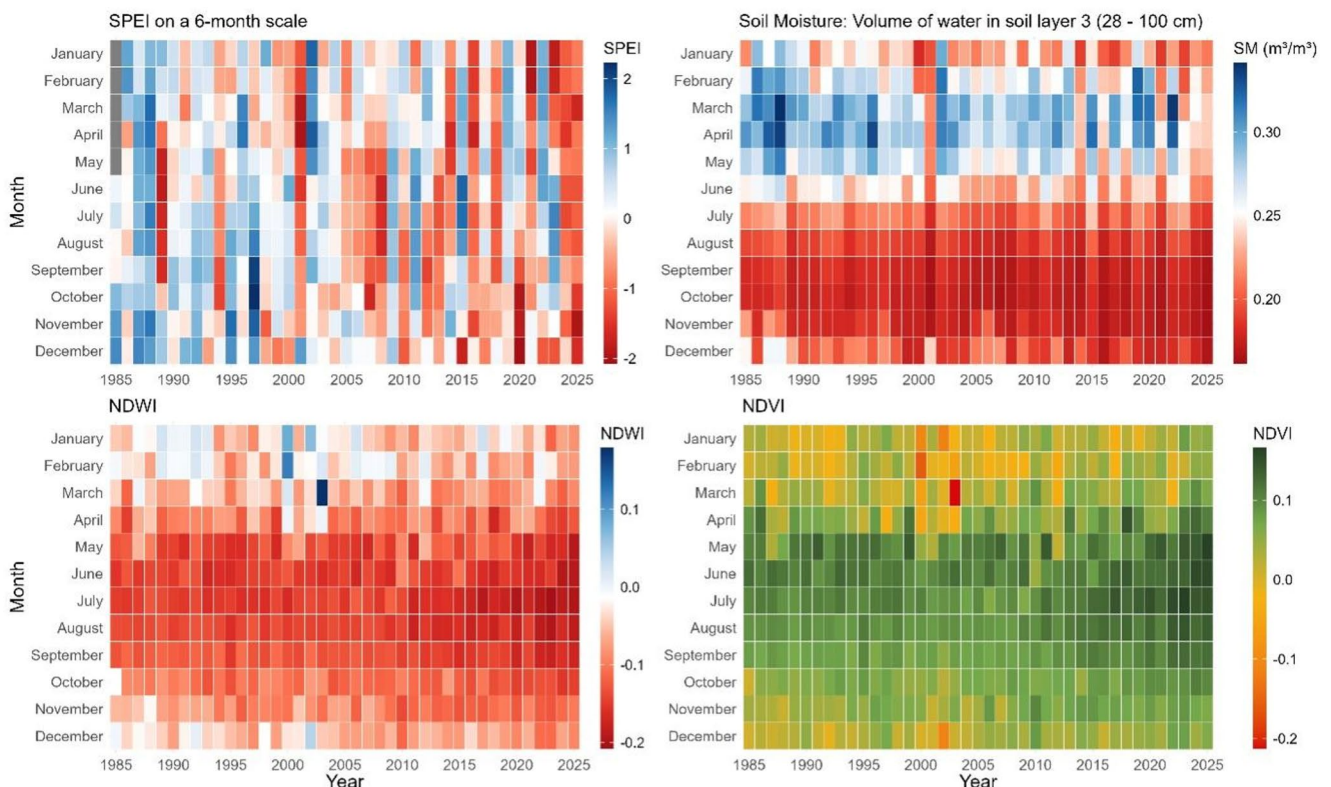


Fig. 3 Heatmaps illustrate the temporal variability of SPEI, soil moisture, NDWI, and NDVI computed from monthly time series for the 1985–2025 period in the KCB

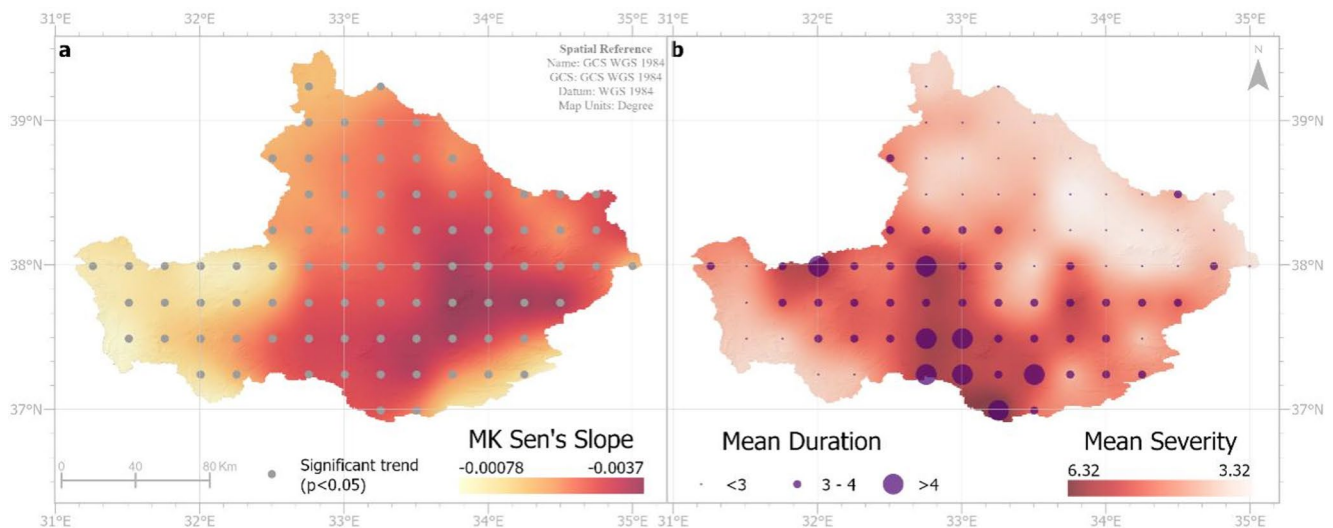


Fig. 4 Spatial patterns of **a** the MK trend test results (Sen's slope) and **b** drought severity and duration derived from SPEI time series for the 1985–2025 period at 81 grid points across the KCB

central and southern parts of the basin. The results of the MK trend test and Sen's slope analyses reveal statistically significant negative trends across all grid points within the basin (Fig. 4a). Spatially, the lowest Sen's slope values are concentrated in the central, southern, and eastern sectors of the basin. Taken together, the temporal and spatial findings indicate that the KCB has been increasingly dominated in recent years by an intensifying and spatially expanding drought regime, characterized by both increasing severity and prolonged duration.

Responses and trends of biophysical indicators

The results reveal that soil moisture and the NDWI, which represents vegetation water status, exhibit largely similar temporal and spatial patterns (Figs. 3 and 5). Throughout the analysis period, the lowest soil moisture values across the basin were predominantly observed between June and December; however, particularly in recent decades, these values have declined further while also extending into January, effectively encompassing all months of the year. Notably, during 2024–2025, soil moisture remained below $0.25 \text{ m}^3/\text{m}^3$ across all months, indicating an unprecedented persistence of moisture deficit conditions. In parallel, the spatial extent of soil moisture stress has increased markedly in recent years, as illustrated in Fig. S2. The NDWI results indicate that values remained mostly below zero throughout the study period, with the lowest values concentrated during the summer months. Nevertheless, recent decades are characterized by a notable intensification of unfavorable NDWI conditions in terms of both duration and severity. The spatial distributions presented in Fig. S3 demonstrate that vegetation water stress is widespread across the basin and has

expanded to larger areas, particularly in recent years. Furthermore, the results of the MK trend test and Sen's slope analysis indicate predominantly significant negative trends in soil moisture (Fig. 5b) and NDWI (Fig. 5c) across the basin, with the strongest and statistically significant negative trends concentrated in the central part of the basin. In contrast, limited areas in the western and northern sectors of the basin exhibit low magnitude but statistically significant positive NDWI trends.

The sensitivity analysis accounting for serial autocorrelation using TFPW showed that the identified trends were highly robust. Trend direction was retained in 99.1–100% of grid cells across all variables, while the proportion of statistically significant trends remained comparable to the original MK results (100% for groundwater, 100% for SPEI-6, 94.6% for soil moisture, 82.0% for NDWI, and 77.5% for NDVI after TFPW). These results indicate that accounting for serial dependence did not materially alter the principal spatial trend patterns identified by the original analysis.

An examination of the temporal variability of NDVI across the basin reveals a pronounced increasing trend throughout the analysis period (Fig. 3). Although the seasonal concentration of NDVI values is primarily confined to the May–September period, recent years indicate an expansion of this interval to encompass April through October. Particularly over the last decades, NDVI values during the summer months have intensified markedly. The spatial distribution maps of NDVI for the 1985–2025 period presented in Fig. S4 demonstrate that the observed increase is not only temporal but also spatial in nature. This increase has become more widespread across the basin in recent years, with NDVI increments reaching approximately 0.2 in the central parts of the basin during the 2018–2025 period, representing

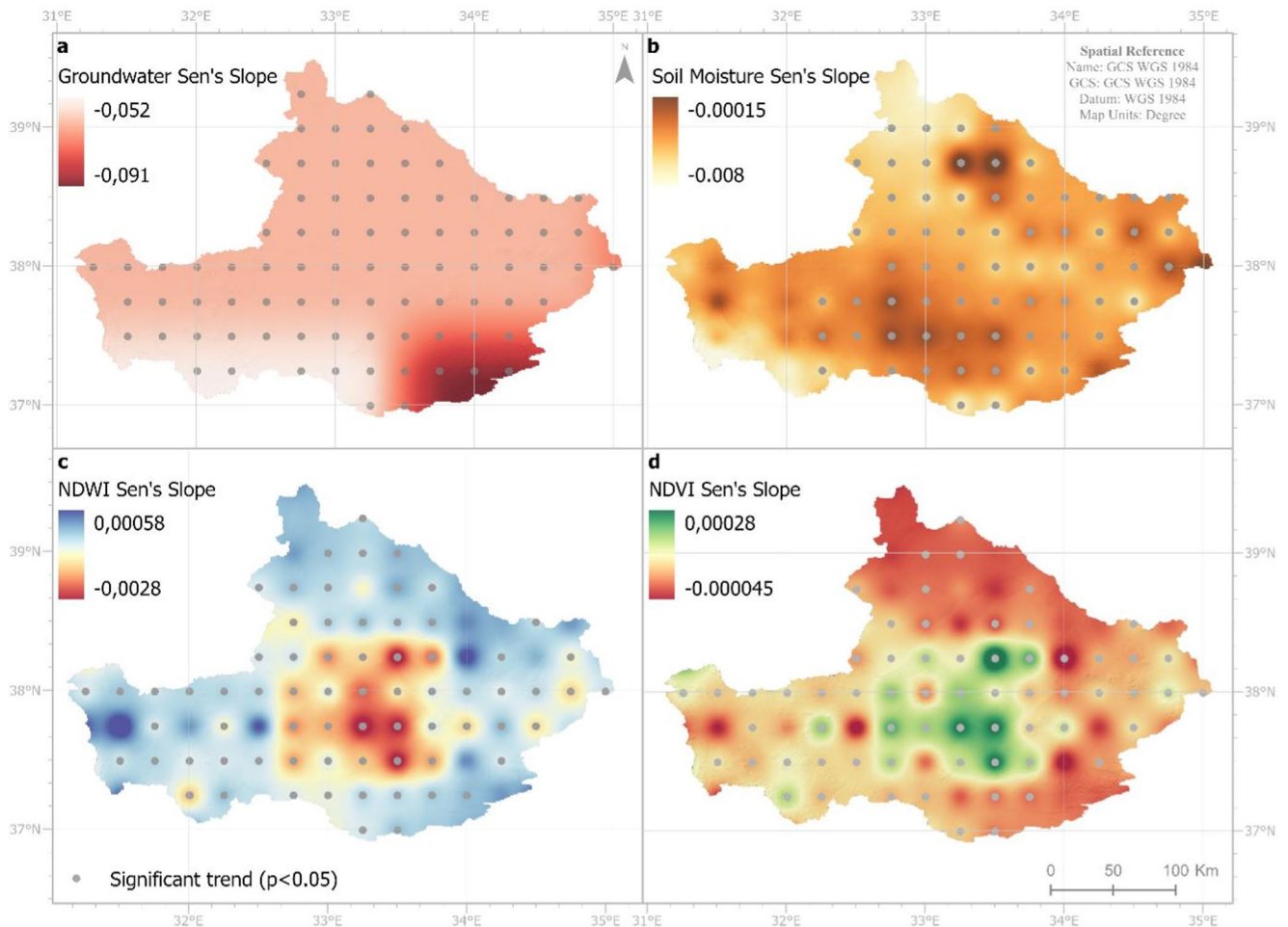


Fig. 5 Spatial distribution of the MK trend test results for groundwater, soil moisture, NDVI, and NDWI derived from 81 grid points across the KCB for the 1985–2025 period

the highest observed values. In addition, the results of the MK trend test and Sen's slope analysis presented in Fig. 5d indicate that predominantly significant positive trends characterize NDVI across the basin. The spatial concentration of the highest positive and statistically significant Sen's slope values in the central sectors of the basin suggests that the observed increase in NDVI reflects a long-term and persistent trend.

Figure 6 illustrates the temporal variability of groundwater level anomalies across the KCB over the 2002–2024 period. The results indicate that groundwater levels generally exhibited positive anomalies until 2015, except for short-term declines observed during certain summer months. However, from 2015 onward, negative anomalies became more frequent and pronounced throughout most of the year. In particular, during the last 5 years of the analysis period, groundwater levels declined by up to approximately 15 cm, with the lowest anomaly values predominantly concentrated between May and November. The spatial distribution maps of groundwater anomalies for

the 2002–2024 period presented in Fig. S5 clearly reveal a marked intensification of groundwater stress across the basin after 2015. This stress has increased both in severity and spatial extent since 2020, spreading across much of the basin. Furthermore, the results of the MK trend test and Sen's slope analysis shown in Fig. 5a indicate statistically significant negative trends at all grid points across the basin. An examination of the spatial distribution of Sen's slope values reveals that the strongest negative trends are concentrated in the eastern parts of the basin.

Spatiotemporal patterns of land use/land cover change

Figure 7 provides an integrated overview of the change detection analyses conducted for four consecutive time intervals (1990–2000, 2000–2006, 2006–2012, and 2012–2018) over the period 1990–2018, based on CORINE LULC data. The Sankey diagram quantitatively depicts areal transitions among LULC classes between successive periods,

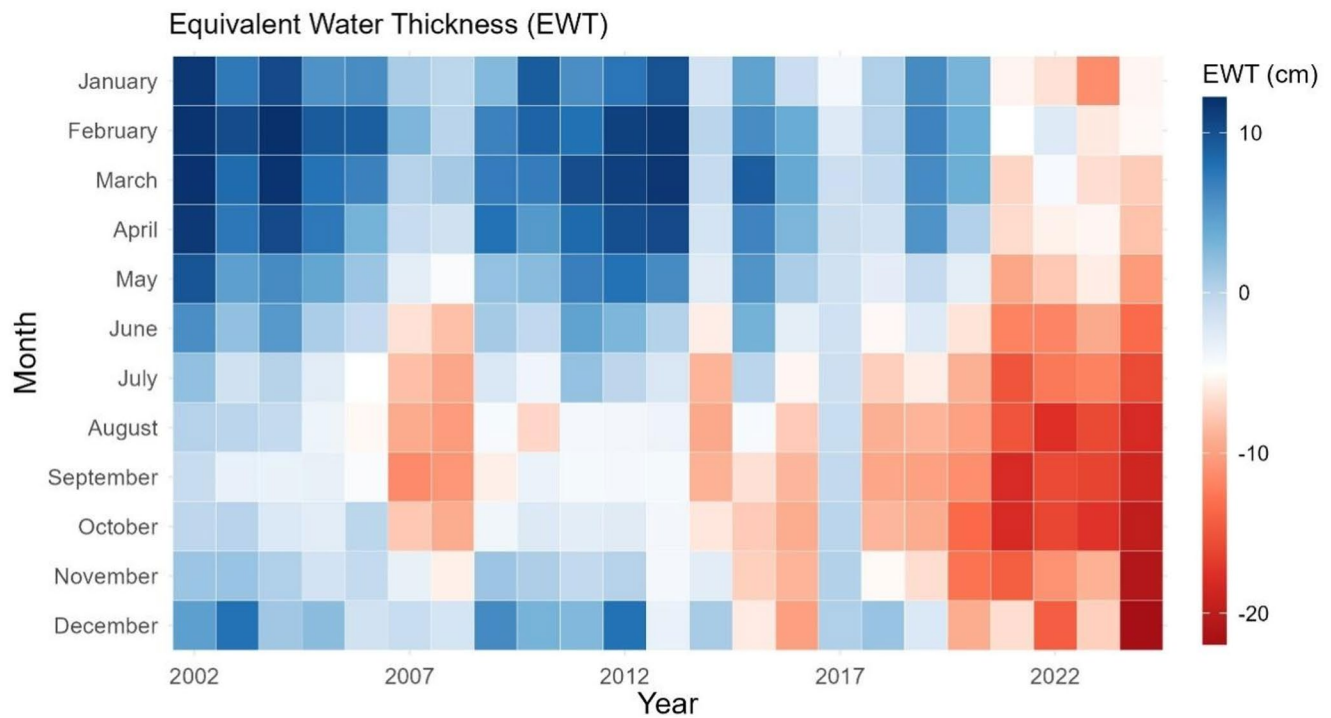


Fig. 6 Heatmap illustrates the temporal variation of groundwater storage anomalies across the KCB for the 2002–2024 period

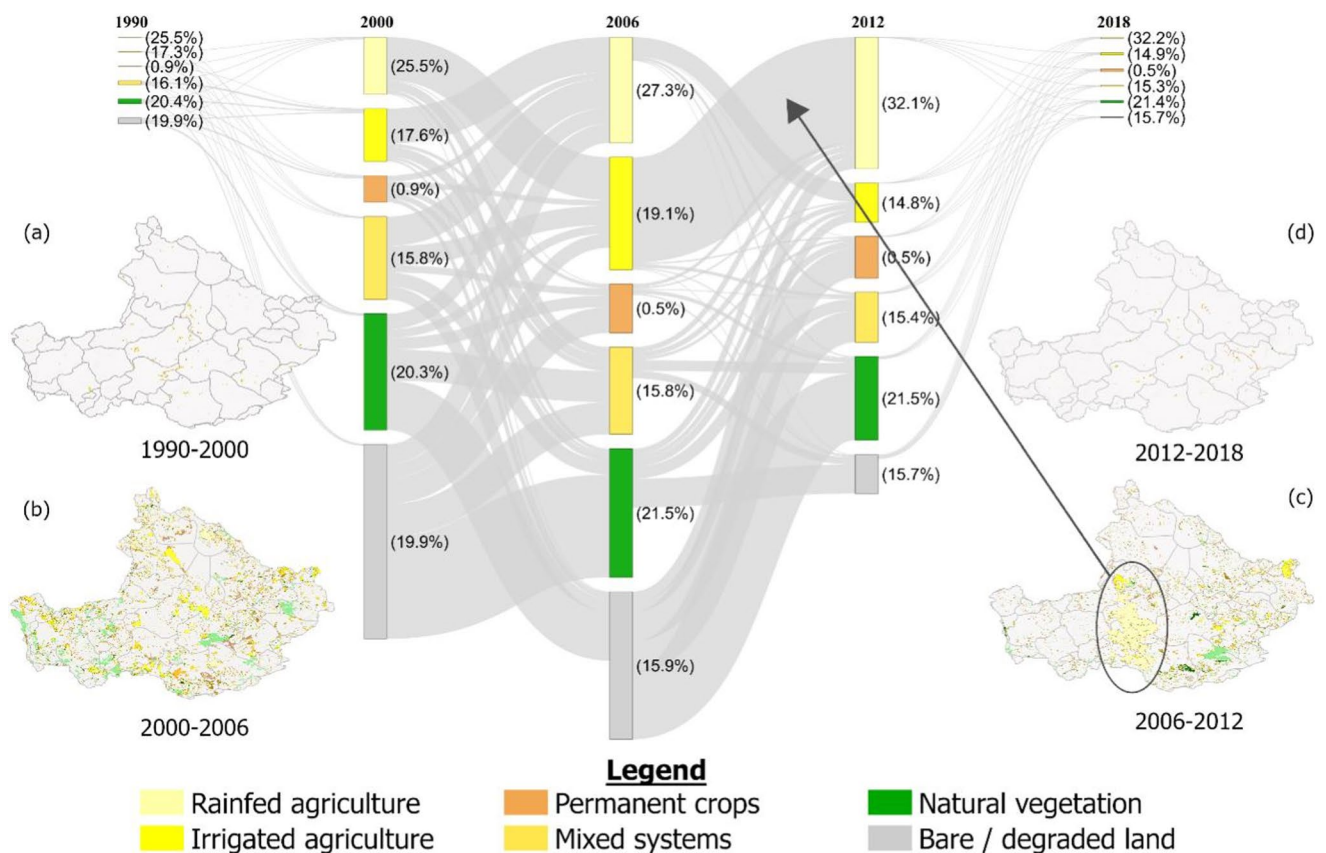


Fig. 7 LULC change detection analysis of the KCB: Spatial distribution maps for the period 1990–2018 and the corresponding Sankey diagram illustrating inter-period land class transitions. While the maps

display spatial patterns across the basin, the Sankey diagram quantifies the trajectory of class conversions and identifies dominant land cover transformation trends

while the maps (Fig. 7a–d) illustrate the spatial distribution of land-use changes occurring within each interval. The percentage values shown in parentheses next to the nodes in the Sankey diagram represent the relative contribution of each class to the total land-use change during the corresponding period. The LULC data presented in the maps reflect only areas undergoing change, rather than class distributions per se. All identified areal (ha) and proportional (%) transition values are summarized in detail in Table S1.

During the 1990–2000 period, land-use patterns exhibited only limited changes, whereas the 2000–2006 interval was characterized by an expansion of agricultural systems and pronounced transformation of natural areas across the KCB. In this period, the transition from natural vegetation to bare/degraded land constituted the dominant conversion process, covering 225,305 ha (19.39%), while the conversion from rainfed agriculture to irrigated agriculture, amounting to 94,478 ha (8.13%), was also notable (Table S1). Concurrently, the conversion of mixed systems to both rainfed agriculture (5.57%) and irrigated agriculture (3.62%) indicates a spatially heterogeneous yet overall intensifying land-use dynamic in agricultural production systems. The land-use changes observed during this interval did not exhibit a distinct spatial concentration; instead, they were distributed across the basin, with comparatively limited changes occurring in the central part of the basin (Fig. 7b).

The 2006–2012 period represents a marked shift in the direction of land-use dynamics. During this interval, the transition from irrigated agriculture to rainfed agriculture emerged as the overwhelmingly dominant conversion process, accounting for 231,248 ha and 27.41% of the total land-use change (Fig. 7). At the same time, increased transitions from bare/degraded land toward natural vegetation (17.33%) and mixed systems (8.09%) indicate a partial recovery trend counteracting the land degradation observed in the preceding period (Table S1). Conversions from irrigated agricultural areas to other land-use classes were found to be largely concentrated in the central part of the basin, where a spatially monolithic pattern of change developed, particularly within the area highlighted by the ellipse in Fig. 7c. This pattern suggests that irrigation-dependent agricultural systems underwent a synchronous and cohesive transformation within a spatially defined core area.

Although land-use changes during the 2012–2018 period occurred over more limited areas, the persistence of bidirectional transitions between bare/degraded land and natural vegetation indicates that the basin-scale system remains dynamic. With respect to agricultural irrigation systems, rainfed and irrigated agricultural lands accounted for 25.5% and 17.3% of the total basin area in 1990, respectively, whereas by 2018 these proportions had shifted to 32.2% and 14.9% (Fig. 7). This overall pattern clearly demonstrates

a basin-wide expansion of rainfed agriculture and a pronounced contraction of irrigated agricultural areas over the analysis period.

Lagged and regime-based dynamics of the socio-ecological system

Figure 8 illustrates the variation of lagged cross-correlation coefficients between SPEI and groundwater storage, soil moisture, NDWI, and NDVI across post-drought lag times ranging from 0 to 24 months. The results reveal pronounced differences among the variables in terms of both correlation magnitude and response lag times (Fig. 8 and Table S2). Statistically significant correlations ($p < 0.05$) are indicated in Fig. 8, allowing the strength and temporal persistence of significant drought-response relationships to be distinguished from non-significant correlation patterns.

The highest positive correlation between groundwater storage and SPEI is observed at zero lag, with a correlation coefficient of approximately $r \approx 0.34$, and this relationship is statistically significant ($p < 0.05$). In subsequent months, the correlation decreases rapidly, reaching negative values at lags of approximately 6–8 months and attaining a minimum of about $r \approx -0.08$. At longer lag times, the correlation coefficients gradually approach zero. The relationship between soil moisture and SPEI exhibits a similar pattern, with the strongest positive correlations occurring at short lag times. The maximum correlation is observed at zero lag, with a coefficient of approximately $r \approx 0.24$. Statistically significant positive correlations are confined to the initial lag period, after which the relationship weakens. Thereafter, the correlation coefficients decline markedly, approaching near-zero or weakly negative values at lags of around 8–10 months.

Cross-correlations between NDWI and SPEI are generally positive, with maximum values occurring at lag times of approximately 3–5 months, reaching $r \approx 0.15$ – 0.16 . At longer lag times, the magnitude of the correlation coefficients decreases, approaching zero, and after the 12th month, only weak positive correlations persist. However, none of the SPEI-NDWI correlations within the 0–24-month lag window are statistically significant ($p > 0.05$).

In contrast, correlations between NDVI and SPEI are predominantly negative across the entire lag range. The lowest correlation coefficients are observed at short lag times, ranging from $r \approx -0.13$ to -0.15 within the 0–4 month interval. At longer lag times, the absolute magnitude of the correlation coefficients decreases, with weak negative values persisting in the 15–24 month range. As with NDWI, no statistically significant lagged correlation between SPEI and NDVI is detected within the examined 0–24-month period ($p > 0.05$). Overall, the maximum and minimum correlation

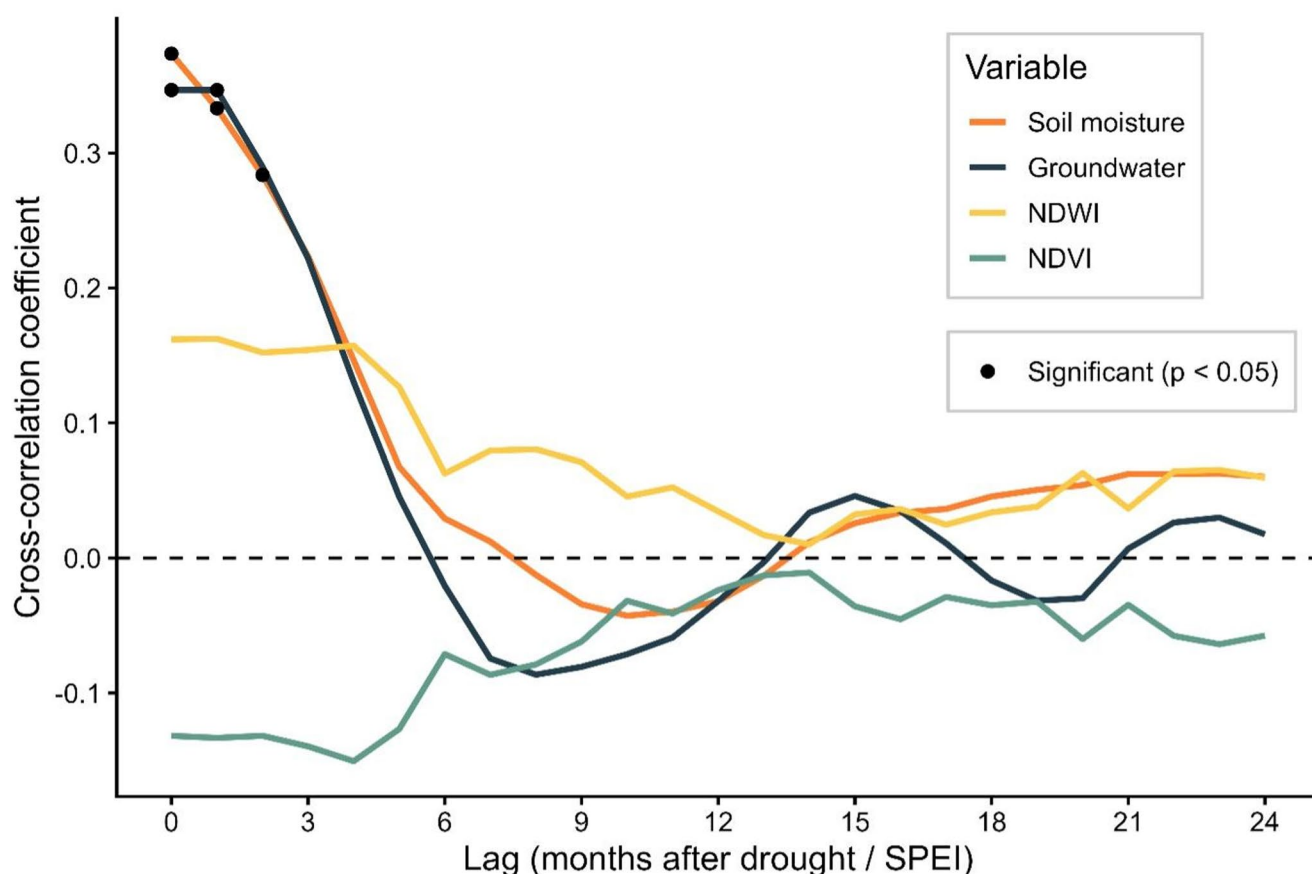


Fig. 8 Variation of cross-correlation coefficients between SPEI and groundwater storage, soil moisture, NDWI, and NDVI across post-drought lag times ranging from 0 to 24 months. The dashed line indicates the zero-correlation level

coefficients and their corresponding lag times for each variable are presented in detail in Table S2.

Figure 9 presents the distributions of standardized anomalies of SPEI, NDWI, NDVI, soil moisture, and groundwater storage in the KCB across the pre-2006, 2006–2012, and post-2012 periods, visualized using ridge plots. The period classification is based on LUCC analyses, which identified substantially higher land conversion rates during 2006–2012 compared to the other periods, particularly transitions from permanently irrigated to non-irrigated areas across the basin. The SPEI distributions indicate that median values were centered around positive z-scores in the pre-2006 period, shifted closer to zero during 2006–2012, and moved markedly toward negative values in the post-2012 period. Similarly, soil moisture and NDWI exhibit a clear shift in the central tendency of their distributions across periods, with predominantly positive anomalies in the pre-2006 period transitioning to distributions concentrated around negative anomalies in the post-2012 period. For these variables, the transition to the post-2012 period is also characterized by a widening of the distributions, increased concentration in the lower tail (negative extremes), and higher interquartile range (IQR) values compared to earlier periods.

In terms of groundwater anomalies, the distribution center is located at positive z-scores in the pre-2006 period, expands toward both positive and negative values during 2006–2012, and shifts below zero in the post-2012 period, where negative anomalies become dominant. A pronounced clustering in the lower tail and increased variance of groundwater anomalies is evident in the post-2012 period. In contrast to the other biophysical indicators, NDVI displays a distinct distributional behavior. While the median NDVI anomalies are close to slightly negative values in the pre-2006 period, the distribution becomes balanced around zero during 2006–2012 and shifts toward positive z-scores in the post-2012 period. The increased concentration in the upper tail and the higher frequency of positive NDVI anomalies during this period stand in clear contrast to the negative shifts observed in the distributions of SPEI, soil moisture, NDWI, and groundwater.

Variable-specific critical thresholds (CT) were identified using univariate breakpoint analysis applied to standardized anomaly (z-score) values of individual time series, with the first statistically significant structural break defined as representing a regime shift. System-level transformations were quantified through breakpoint analysis performed on the

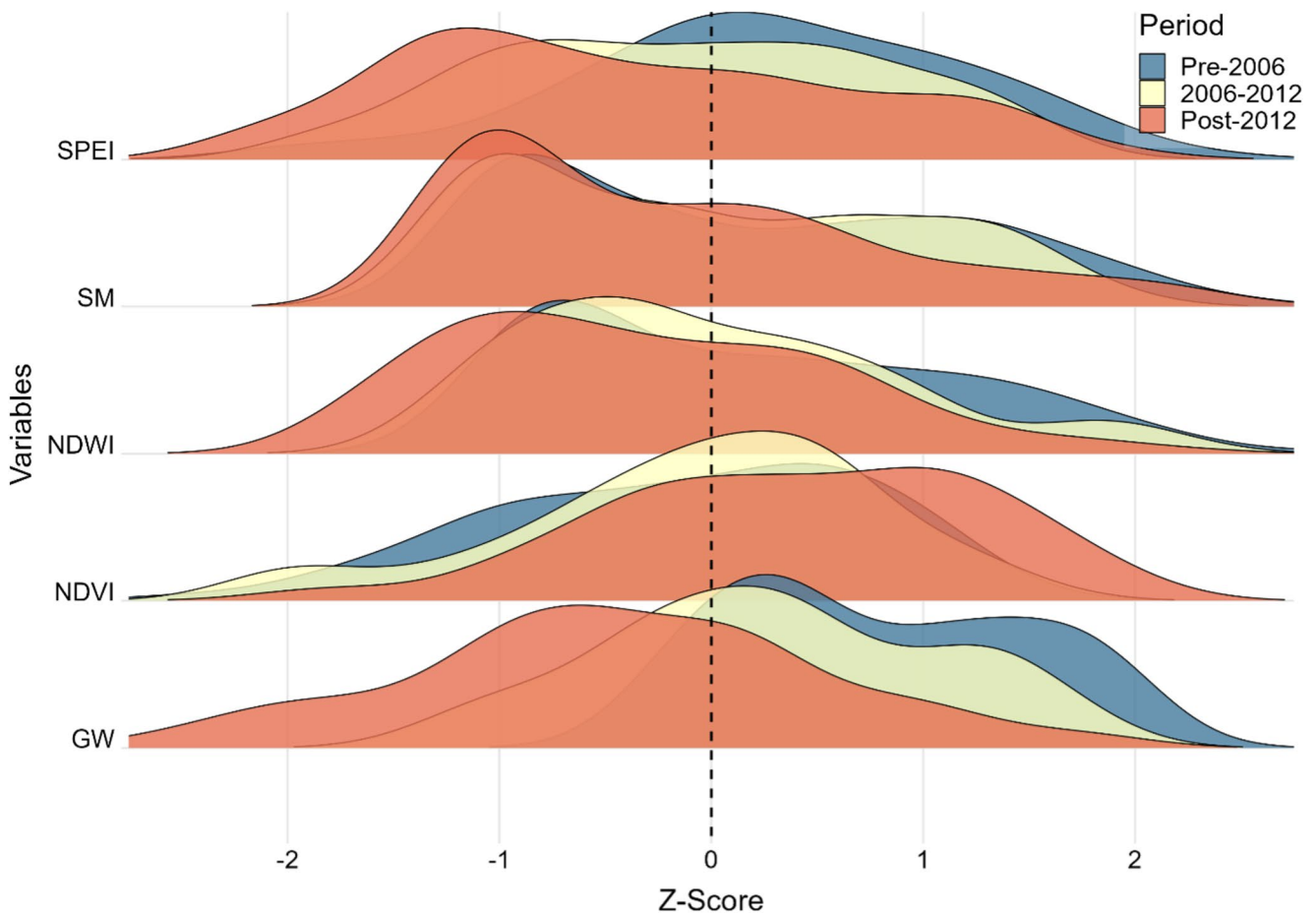


Fig. 9 Multivariate regimes shift analysis of SPEI, groundwater, soil moisture, NDWI, and NDVI variables: **a** Ridge plots displaying the standardized anomaly (z-score) distributions across three distinct peri-

ods: pre-2006, 2006–2012, and post-2012. The abbreviations SM and GW stand for soil moisture and groundwater, respectively

time series of the first principal component (PC1) derived from a multivariate standardized anomaly matrix. Within this framework, two alternative system representations were evaluated: (i) a PCA structure excluding groundwater to preserve temporal continuity and covering the period 1985–2025, and (ii) a PCA structure including groundwater to represent surface-subsurface interactions, constrained to the common data availability period of 2002–2024.

Figure 10 illustrates the standardized anomaly time series of SPEI, NDVI, NDWI, soil moisture, and groundwater storage for a representative grid point characterized by intensive land-use change, particularly a pronounced transition from permanently irrigated agriculture to rainfed agriculture (see Fig. 1). A critical threshold was detected in the SPEI series in November 2004, after which anomaly values predominantly shifted toward negative conditions and the lower tail of the distribution expanded markedly. Groundwater anomalies exhibited a delayed regime shift in May 2006; following this transition, median z-score values persistently shifted to negative levels, with an increasing frequency of

values below -2.0 in later periods. For the vegetation and surface water indicators NDVI and NDWI, synchronous critical thresholds were identified in March 2008. After this point, NDWI anomalies were largely characterized by negative values, whereas the NDVI distribution displayed positive median values accompanied by a broader variance. The root-zone soil moisture series (28–100 cm) yielded a critical threshold in June 1993, after which a progressive widening of the anomaly distribution and increasing variability were observed over time. When multivariate system behavior was evaluated, the PCA structure excluding groundwater indicated a system-level transition in March 2008, with PC1 loadings revealing dominant contributions from NDVI (-0.59), NDWI (0.60), and soil moisture (0.51). In contrast, the PCA structure including groundwater identified a system transformation in April 2006, with PC1 loadings more evenly distributed among groundwater (-0.41), NDVI (0.52), NDWI (-0.52), soil moisture (-0.42), and SPEI (-0.23). Following this transition, the system became

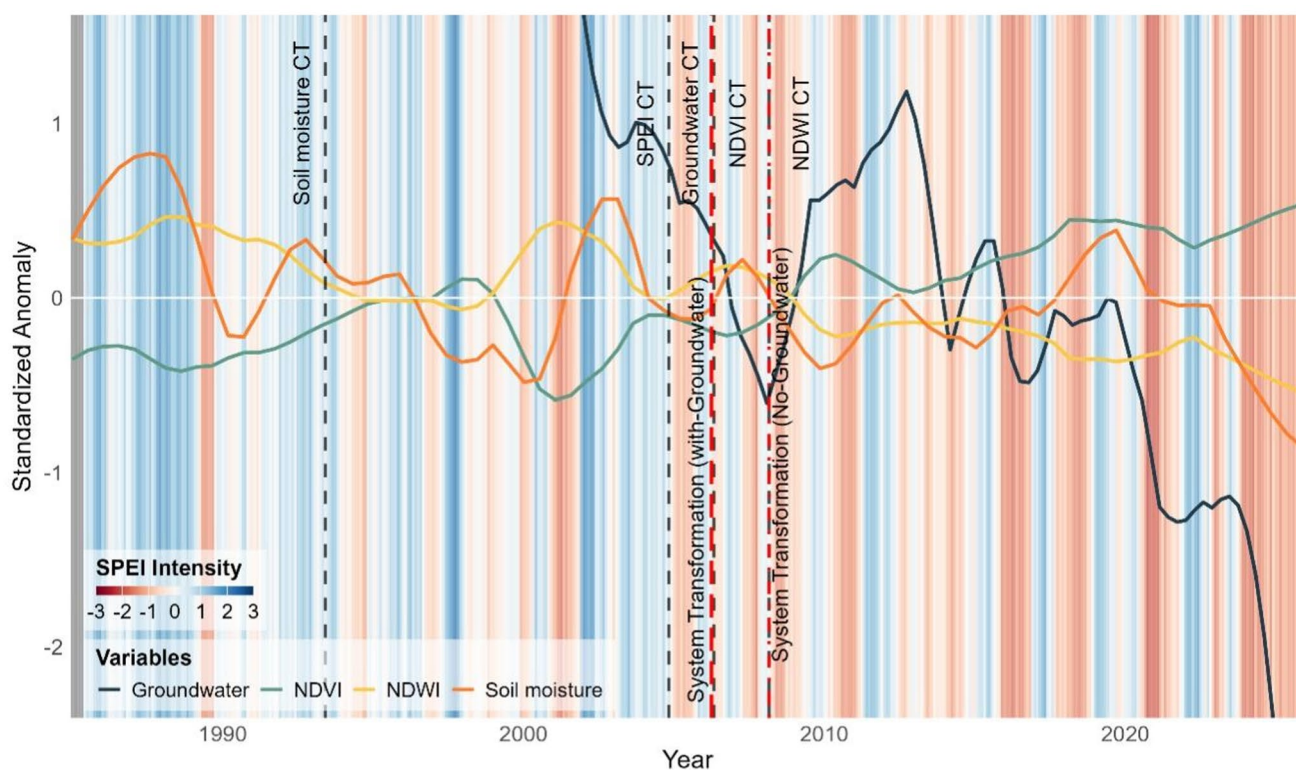


Fig. 10 Temporal evolution of climate hazard and biophysical variables and identified system regime shifts. The figure illustrates the synchronization between SPEI-defined drought periods (climate stripes) and negative anomalies in groundwater and vegetation indices. The time series corresponds to a representative grid point selected from an

area exhibiting pronounced land-use change, specifically a transition from continuously irrigated to rainfed agricultural systems. The spatial location of the selected grid point is shown in Fig. 2. Vertical markers indicate key transition points associated with structural changes in the ecosystem. CT stands for critical threshold

characterized by persistent negative groundwater anomalies concurrent with ongoing drought conditions (Fig. 10).

Except for the soil moisture critical threshold (1993), the critical thresholds identified for SPEI (2004), groundwater (2006), and NDVI-NDWI (2008), along with both system-level transformations (2006 and 2008), are concentrated within the 2006–2012 period, during which the most pronounced land use changes were recorded across the basin. This interval coincides with the phase in which transitions from irrigated to non-irrigated agricultural land were most extensive, as illustrated by the Sankey diagram and land use maps (Fig. 7).

Discussion

SPEI analyses indicate a pronounced increasing trend in both the duration and severity of drought across the KCB. This trend has become particularly evident since the mid-2000s and has expanded spatially throughout the basin. The findings are highly consistent with previous basin-scale studies (Kartal and Nones 2024; Serkendiz et al. 2023, 2024, 2025; Serkendiz and Tatli 2023), suggesting that meteorological

drought has transitioned from a transient climatic stressor to a persistent perturbation that actively restructures the functioning of the agricultural SES. This interpretation is supported by the observation that the strongest positive correlations between SPEI and both groundwater storage and soil moisture occur at zero lag ($r \approx 0.34$ and $r \approx 0.24$, respectively) and weaken rapidly thereafter, indicating a rapid yet unsustainable transmission of meteorological drought signals into basin hydrology. Furthermore, critical threshold analysis reveals a persistent shift of SPEI anomalies toward negative values after 2004, accompanied by a pronounced concentration in the lower tail of the distribution. Collectively, these results demonstrate that drought should be conceptualized not merely as a climatic hazard, but as an internally reinforcing process that triggers and amplifies system-level interactions. In this context, Roy et al. (2025) emphasize the decisive role of human interventions in the propagation of meteorological drought into hydrological and agricultural drought types, framing drought as a multi-component and cascading process.

From a SES perspective, it is evident that the drought impacts observed in the basin are non-linear and are amplified through multiple interacting feedback mechanisms.

This non-linear behavior can be explained by threshold dynamics, whereby the accumulation of slow-changing variables over time may trigger abrupt regime shifts (Lambin et al. 2003; Li et al. 2020). The causal loop diagram (CLD) presented in Fig. 11 conceptually illustrates this complex feedback structure. Among the most dominant reinforcing feedbacks identified in the diagram is the interaction between drought and groundwater (R1). Increasing drought frequency and severity constrain natural groundwater recharge; this effect is further intensified by the closed-basin setting and the dominance of precipitation as the primary recharge source. The rapid transition of the SPEI-groundwater correlation to negative values at short lags (6–8 months; $r \approx -0.08$) indicates that drought impacts accumulate over time, driving the hydrological system into a self-reinforcing degradation cycle. Declining groundwater levels, in turn, deepen hydrological drought and amplify the effects of meteorological drought, giving rise to a self-perpetuating “drought trap.” In a similar vein, Roy et al. (2022)

identify groundwater abstraction as one of the most critical components of vulnerability to agricultural drought.

Another major reinforcing feedback triggered by drought operates through declines in soil moisture (R2). The analyses indicate that drought conditions substantially reduce soil moisture, which in turn increases plant water stress and stimulates irrigation demand. Indeed, the strongest relationship between SPEI and soil moisture occurring at zero lag ($r \approx 0.24$), followed by a weakening of the association within 8–10 months, demonstrates that meteorological drought signals are rapidly transmitted to agricultural systems. Rising irrigation demand subsequently accelerates groundwater abstraction, further intensifying drought conditions. This feedback loop is corroborated by the negative trends observed in soil moisture and NDWI indicators, particularly over the past decades. As shown in Fig. 9, the pronounced shift of soil moisture and NDWI anomalies toward negative values in the post-2012 period, together with increased lower-tail concentration and expanding IQRs, indicates that this process has evolved into a structural regime shift.

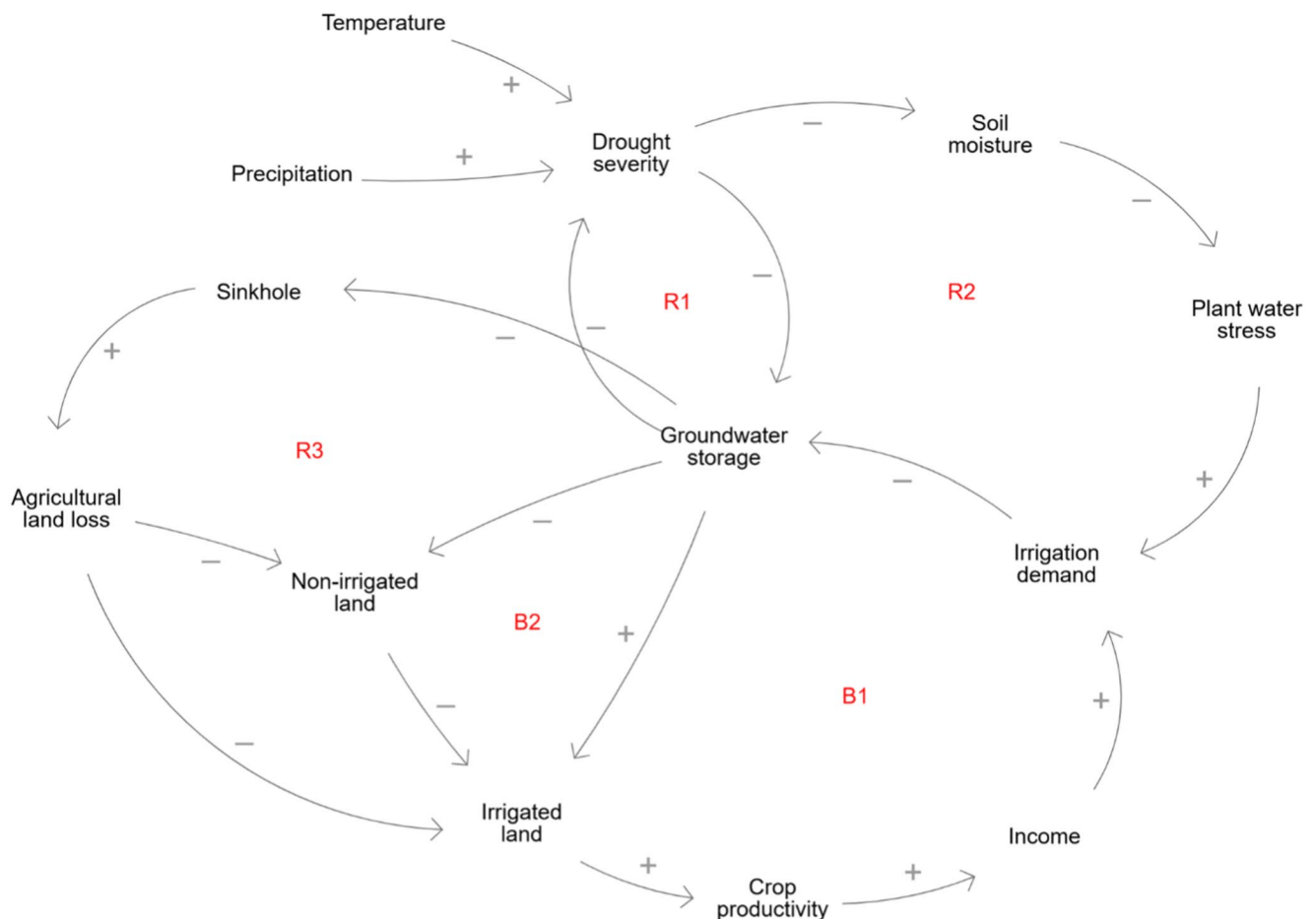


Fig. 11 Causal loop diagram illustrating the nonlinear interactions of drought perturbations within an agricultural socio-ecological system in the KCB, highlighting the linkages between ecological and social subsystems. In the diagram, the loops labeled “R” and “B” represent the

reinforcing and balancing feedback mechanisms, respectively, while the “+” and “-” symbols on the arrows indicate positive and negative relationships between variables

The growing temporal and spatial coherence of declining soil moisture and increasing vegetation water stress during the irrigation season suggests that this process is no longer transient but structural in nature. The statistically significant negative MK trends identified in areas dominated by irrigated agriculture further indicate that agricultural water use has assumed a reinforcing rather than a mitigating role with respect to drought. These findings are consistent with the assessments of cascading drought impacts reported by Serkendiz et al. (2025).

Increasing irrigation activities promote the expansion of irrigated agricultural areas by generating short-term gains in agricultural production and income. However, this process simultaneously intensifies pressure on groundwater resources, introducing a balancing mechanism that constrains the system through resource limitations (B1). The increase in the share of irrigated agriculture from 17.6 to 19.1% during the 2000–2006 period, together with the positive NDVI anomalies observed over the same interval, indicates a strengthening of irrigation during this phase. Nevertheless, the consistently negative relationship between NDVI and SPEI across the entire lag range ($r \approx -0.13$ to -0.15) suggests that vegetation enhancement has been sustained primarily by anthropogenic irrigation interventions rather than by climatic improvement. As emphasized by Lambin et al. (2003), such expansion should be interpreted not only as a response to biophysical conditions but also as the outcome of underlying socio-economic drivers, including institutional investments, market dynamics, and policy choices. However, Falloon and Betts (2010) caution that increasing dependence on irrigation in semi-arid regions may heighten long-term agricultural vulnerability. In this context, it can be argued that the growing water demand in the KCB has increasingly pushed groundwater reserves toward their resilience thresholds, rendering the observed expansion unsustainable in the long term.

Indeed, the land use changes observed during the 2006–2012 period provide clear indications that this threshold had been exceeded. During this interval, approximately 231,248 ha of irrigated agricultural land transitioned to non-irrigated classes, increasing the share of rainfed areas in the basin from 27.3 to 32.1%, while the proportion of irrigated land declined from 19.1 to 14.8%. This transformation closely coincides with the regime shift in groundwater anomalies, which permanently shifted to negative values after 2006 (CT: 2006). As illustrated in Figs. 9 and 10, the pronounced transition to negative groundwater anomalies and the sharp decline in groundwater levels during the same period indicate that declining groundwater availability was the primary driver of this land use conversion. Similarly, Kartal and Nones (2024) and Serkendiz et al. (2025) associate the post-2008 land use changes in the basin largely

with groundwater depletion. More broadly, multi-temporal assessments of land-use transitions provide an important basis for identifying the direction and intensity of landscape reorganization under changing environmental conditions (Dutta et al. 2026). Taken together, these findings are consistent with the assessment of Lankford et al. (2023), who emphasize the limited capacity of both irrigations return flows and precipitation to recharge groundwater in semi-arid basins. While increasing groundwater abstraction in the basin has delivered short-term production gains, it has ultimately evolved into a balancing mechanism that constrains the system through resource depletion, thereby eroding long-term productive capacity (B2).

The concurrent effects of drought, intensive irrigation, and declining groundwater levels collectively erode the natural replenishment capacity of groundwater, thereby increasing the basin's susceptibility to geomorphological hazards. Numerous studies have demonstrated that the KCB exhibits a high sensitivity to sinkhole formation (Orhan et al. 2020, 2024; Sarı et al. 2021). Intensive irrigated agriculture is widely recognized as one of the primary triggers of sinkhole development in the basin. The clustering of groundwater anomalies in the lower tail and the marked increase in variance during the post-2012 period indicate that these physical risks have become systemic and persistent. The linkage between groundwater depletion and sinkhole formation constitutes not only a physical hazard but also generates an additional reinforcing feedback process (R3) by amplifying socio-economic vulnerability through agricultural land losses and declining farm incomes. Such cascading effects demonstrate how an external shock (drought) can initiate multi-layered degradation loops across the entire system, as documented in different socio-ecological contexts by Murungweni et al. (2011) and Sanga et al. (2021).

Overall, the temporal and spatial analyses based on SPEI, soil moisture, NDWI, and groundwater indicators reveal that the intensification of drought has generated multi-layered reinforcing feedback loops between the biophysical and socio-economic components of the basin, thereby amplifying drought impacts. Land-use changes and groundwater depletion further exacerbate drought effects through their influence on farmers' incomes and agricultural production capacity. The shift of NDVI anomalies toward positive values during the post-2012 period, in contrast to the transition of NDWI, soil moisture, groundwater, and SPEI indicators into a persistent negative regime (Fig. 9), indicates the adoption of maladaptive practices that prioritize the maintenance of existing production systems rather than enhancing adaptive capacity to climatic stress (Serkendiz et al. 2025). This pattern suggests that short-term adaptation strategies in the basin may increase socio-ecological vulnerability over the long term.

Integrating LULC dynamics with remotely sensed environmental indicators can provide important insights into the environmental responses associated with landscape transformation. For example, Choudhury et al. (2023) demonstrated the value of jointly evaluating multi-temporal LULC changes and NDVI alongside other environmental indicators to identify relationships between landscape transformation and environmental conditions.

Despite intensifying drought conditions, declining groundwater levels, and pronounced land-use transformations, the apparent increase in NDVI values suggests an artificial greening signal, indicating that the basin's socio-ecological system is evolving toward an increasing regime of uncertainty rather than maintaining a linear and stable equilibrium. This pattern implies a transition toward a complex adaptive system characterized by nonlinearity and emergent behavior. Such dynamics are fully consistent with the SES framework, which conceptualizes humans and nature not as independent entities but as intertwined and mutually constitutive components (Folke 2006). In this context, human activities, particularly water use and land management decisions, are reshaping ecosystem dynamics from local to basin scales, thereby progressively undermining the system's capacity to sustain ecosystem services.

The exceedance of critical threshold values by SPEI in 2004, groundwater levels in 2006, and the NDWI and NDVI indicators in 2008 provide strong statistical evidence that the observed system transformation intensified around 2008. Such threshold crossings indicate, in line with Holling's (1973) concept of ecological resilience, that the system does not operate around a single equilibrium state but rather encompasses multiple potential stability domains, within which abrupt regime shifts may occur once disturbance levels surpass critical limits. In particular, the progressive accumulation of long-term pressures on slow variables such as groundwater leading to the attainment of critical thresholds constitutes a characteristic manifestation of nonlinear socio-ecological system behavior, as emphasized by Li et al. (2020).

In this context, the large-scale land use changes that occurred during the 2006–2012 period reflect not only a spatial transformation but also a reconfiguration of feedback mechanisms between the social and ecological subsystems. The increasing dominance of negative feedback loops suggests a weakening of the stabilizing mechanisms that previously maintained the system in its “normal” state, thereby pushing the system toward an alternative stability domain. Based on the hypothetical conceptual diagram presented in Fig. 12, this process can be interpreted as the system approaching a critical threshold and transitioning from a normal state (Fig. 12a) to an alternative regime (Fig. 12b).

However, as emphasized in the SES literature, a transition to an alternative regime does not necessarily imply a complete system collapse (Folke 2006). Despite increasing drought pressure and critical declines in groundwater levels, the continuation of agricultural production indicates that the system remains within a certain range of resilience. When interpreted within the adaptive cycle framework, this suggests that the system is becoming increasingly rigid within the “conservation” phase yet has not fully entered the “release” phase. Nevertheless, as highlighted by Ostrom (2009), if the interactions among social, ecological, and institutional variables operating across different spatial and temporal scales are not adequately understood, such fragile equilibrium states may rapidly evolve into abrupt and potentially irreversible regime shifts.

At this stage, the decisive role of governance and management choices in shaping system dynamics becomes particularly evident. If maladaptive practices are replaced by adaptation-oriented strategies, such as the adoption of water-sensitive irrigation techniques, a shift toward crop patterns with lower water demand, effective groundwater management, and enhanced farmer awareness, the system may be steered away from a critical threshold that could otherwise lead to irreversible collapse (Fig. 12c). Such interventions are consistent with the concept of adaptive governance articulated by Folke (2006), which emphasizes the coordinated functioning of diverse knowledge systems, social networks, and multi-scale institutional arrangements.

Sustainable groundwater governance in the KCB therefore requires moving beyond general water-conservation measures toward an integrated management approach that combines enforceable abstraction controls with participatory and adaptive governance. In particular, effective metering and monitoring of groundwater withdrawals, clearly defined abstraction limits, and credible enforcement mechanisms are essential where increasing irrigation demand places persistent pressure on aquifer reserves (Ross and Martinez-Santos 2010). These regulatory measures should be complemented by co-management arrangements that actively involve farmers and groundwater-user groups in rule development, monitoring, and implementation, while maintaining strong institutional oversight and transparency (Hoogesteger 2022). Evidence from the High Plains aquifer further demonstrates that flexible multi-year water allocations, locally adapted conservation targets, and the integration of bottom-up rule development with top-down enforcement can substantially reduce groundwater use and aquifer depletion without necessarily compromising farm incomes (Orduña Alegría et al. 2024). For the KCB, such groundwater governance should be integrated with drought-adaptation measures, including water-efficient irrigation, shifts toward less water-intensive crops, and

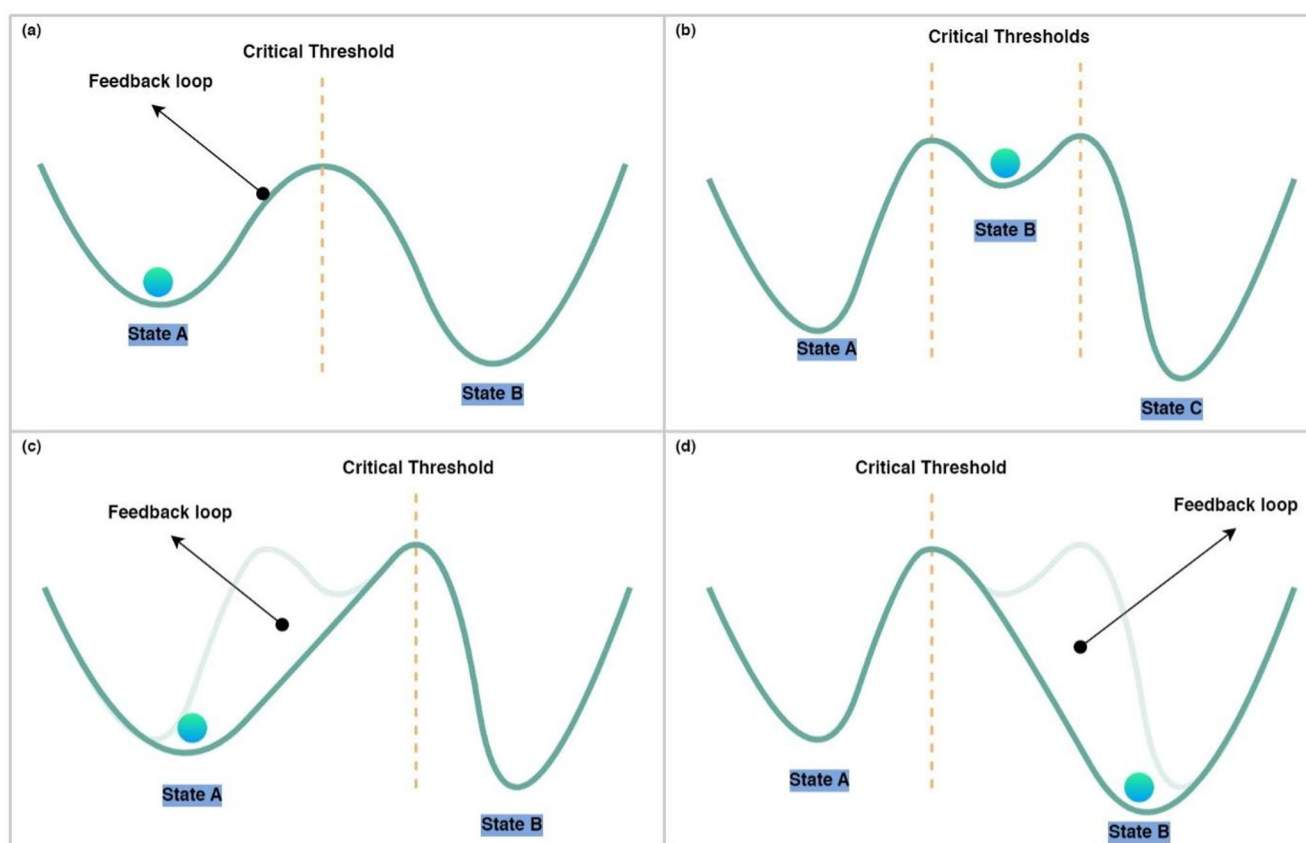


Fig. 12 Hypothetical conceptual stability landscape illustrating alternative system states, feedback mechanisms, and critical thresholds in a socio-ecological system under increasing drought pressure and human intervention. Panel (a) represents the normal system state (State A), where stabilizing feedbacks maintain system resilience within a single basin of attraction. Panel (b) illustrates a regime shift to an alternative stable state (State B) after crossing a critical threshold, driven by dominant negative feedbacks associated with prolonged drought,

groundwater depletion, and land-use change. Panel (c) depicts a potential recovery pathway in which adaptive management and governance interventions reshape feedbacks, allowing the system to move away from a critical threshold and regain resilience. Panel (d) shows a maladaptive trajectory where continued unsustainable practices reinforce destabilizing feedbacks, pushing the system beyond resilience limits toward an irreversible degraded state. Dashed lines indicate critical thresholds separating alternative stability domains

drought-responsive adjustment of groundwater allocations. Collectively, these measures could help redirect adaptation from short-term reliance on groundwater abstraction toward a more sustainable strategy that maintains agricultural livelihoods while safeguarding long-term aquifer resilience.

Conversely, if current maladaptive practices persist and groundwater levels continue to decline, the system may cross a critical threshold and move beyond its resilience thresholds (Fig. 12d). Such a scenario could reinforce the drought-food insecurity feedback loops highlighted by Roy et al. (2025), thereby exacerbating negative interactions among agricultural production, water resources, and social well-being. Ultimately, the agricultural socio-ecological structure of the basin may become irreversibly degraded, entailing not only an increased risk of ecological collapse but also escalating systemic risks, most notably to food security, along with heightened social uncertainty.

Although developed and evaluated within the KCB, the proposed socio-ecological resilience framework is

conceptually transferable to other semi-arid agricultural basins where drought, irrigation dependence, groundwater abstraction, and land-use dynamics interact through comparable feedback mechanisms. Its principal components -climatic perturbation, biophysical responses, human adaptation, resource feedbacks, and potential threshold behavior-provide a general structure for examining how short-term adaptation may generate longer-term vulnerability. However, transferability should not imply direct replication of the thresholds or feedback strengths identified in the KCB. Their magnitude and timing are likely to vary according to basin-specific hydroclimatic conditions, groundwater recharge and storage characteristics, cropping systems, irrigation dependence, and institutional and governance arrangements. Accordingly, applications elsewhere would require locally calibrated indicators and thresholds while retaining the broader feedback-based SES structure.

Conclusions

This study presents a comprehensive framework to examine how increasing drought conditions, groundwater-dependent irrigated agriculture, and rapid LULC changes have reconfigured the dynamics of an agricultural SES in the KCB. Analyses based on remote sensing products, reanalysis data, and long-term time series clearly demonstrate that drought in the basin has evolved from a transient climatic stressor into a persistent perturbation that fundamentally alters system functioning.

The SPEI analyses reveal a marked increase in both the severity and duration of drought since the mid-2000s, with this climatic pressure being rapidly transmitted to agricultural and hydrological systems through indicators such as soil moisture, NDWI, and groundwater storage. In contrast, the positive anomalies observed in NDVI values during the post-2012 period indicate that vegetation productivity has been sustained beyond natural hydroclimatic limits through intensive irrigation and human interventions. This divergence constitutes a clear example of maladaptation; whereby short-term production gains are achieved at the expense of long-term system resilience.

Temporal analyses identify critical thresholds in SPEI (2004), groundwater storage (2006), and NDWI-NDVI (2008), indicating that the system transformation intensified around 2008. These threshold behaviors are primarily associated with the cumulative effects of prolonged pressures on slow-changing variables such as groundwater, which tend to produce nonlinear responses over time. The large-scale LULC transitions observed between 2006 and 2012, particularly shifts from irrigated to rainfed agriculture, closely coincide with groundwater depletion and suggest that the system has been moving toward an alternative stability domain.

CLD further demonstrate that interactions among drought, groundwater abstraction, irrigation practices, and land use are driving the system into a reinforcing degradation process through multi-layered feedback mechanisms. Reinforcing feedbacks amplify the hydrological and agricultural impacts of drought, while balancing mechanisms that temporarily sustain production ultimately increase system vulnerability through resource constraints. This indicates that, although the basin has not yet reached a complete collapse, it currently persists in a fragile alternative regime within its resilience limits.

Overall, the socio-ecological transformation observed in the KCB reflects a complex process jointly shaped by climatic pressures and human-driven water and land management decisions. Without the implementation of strategies such as limiting groundwater abstraction, adopting adaptation-oriented water management, and transitioning toward

less water-intensive agricultural production patterns, the system is likely to cross a critical threshold into a regime from which recovery would be difficult. This study underscores that the impacts of drought on agricultural SESs should be assessed not solely in terms of production losses, but also through the lenses of resilience, feedback mechanisms, and threshold dynamics, all of which are essential for informing sustainable management policies.

Nevertheless, the study has several limitations. As the analyses are primarily based on biophysical indicators, social, governance, and economic dynamics could only be addressed to a limited extent. In particular, farmers' perceptions of drought, adaptation and maladaptation decisions, and water-use behaviors, and how these human dimensions are integrated into system dynamics, represent important avenues for future research. Moreover, the spatial heterogeneity and decision-making processes underlying the large-scale LULC transformations observed between 2006 and 2012 require more detailed investigation at higher spatial and temporal resolutions. In this context, the study provides an initial framework for identifying early warning signals of drought-induced socio-ecological transformations, while clearly highlighting the need for more integrated and interdisciplinary research to better understand long-term system sustainability.

Supplementary Information The online version contains supplementary material available at <https://doi.org/10.1007/s11625-026-01910-z>.

Acknowledgments I dedicate this study to the cherished memory of Professor Telat Koç, who inspired in me a love of science and geography and taught me to understand life as an interconnected whole. He was not merely a scientist, but also an enlightened intellectual who championed solidarity and a culture of coexistence and firmly believed in the universal ideals of humanity. Although he is no longer among us, the values he embraced continue to illuminate my path. I bow in respect before his memory and his legacy.

Author contributions Hıdır Serkendiz: conceptualized the study, designed the methodology, performed the data analysis, interpreted the results, and wrote the original manuscript. The author reviewed and approved the final version of the manuscript.

Funding This research did not receive any specific grant from funding agencies in the public, commercial, or not-for-profit sectors.

Data availability All datasets used in this study are openly available from public repositories. Groundwater storage anomalies were derived from the GRACE and GRACE-FO JPL Global Mascon solutions, obtained from the NASA Jet Propulsion Laboratory (https://grace.jpl.nasa.gov/data/get-data/jpl_global_mascons/). NDVI and NDWI indices were calculated using Landsat satellite imagery provided by the U.S. Geological Survey (USGS) Earth Resources Observation and Science (EROS) Center (<https://www.usgs.gov/landsat-missions/data>). Precipitation, temperature, and soil moisture, were retrieved from the Copernicus Climate Data Store (<https://cds.climate.copernicus.eu/datasets>). LULC data were obtained from the CORINE Land Cover

database provided by the Copernicus Land Monitoring Service (<https://land.copernicus.eu/en/products/corine-land-cover>).

Declarations

Competing interest The author declares that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Ethical approval and consent to participate This study does not involve human participants, human data, or animals. Therefore, ethics approval and consent to participate are not applicable.

Consent for publication Not applicable.

Open Access This article is licensed under a Creative Commons Attribution 4.0 International License, which permits use, sharing, adaptation, distribution and reproduction in any medium or format, as long as you give appropriate credit to the original author(s) and the source, provide a link to the Creative Commons licence, and indicate if changes were made. The images or other third party material in this article are included in the article's Creative Commons licence, unless indicated otherwise in a credit line to the material. If material is not included in the article's Creative Commons licence and your intended use is not permitted by statutory regulation or exceeds the permitted use, you will need to obtain permission directly from the copyright holder. To view a copy of this licence, visit <http://creativecommons.org/licenses/by/4.0/>.

References

- Antoni C, Huber-Sannwald E, Reyes Hernández H, van't Hooft A, Schoon M (2019) Socio-ecological dynamics of a tropical agricultural region: Historical analysis of system change and opportunities. *Land Use Policy* 81:346–359. <https://doi.org/10.1016/j.landusepol.2018.10.028>
- Chen Y, Yu P, Wang L, Chen Y, Chan EHW (2024) The impact of rice-crayfish field on socio-ecological system in traditional farming areas: Implications for sustainable agricultural landscape transformation. *J Clean Prod* 434:139625. <https://doi.org/10.1016/j.jclepro.2023.139625>
- Cox M (2014) Applying a Social-Ecological System Framework to the Study of the Taos Valley Irrigation System. *Hum Ecol* 42(2):311–324. <https://doi.org/10.1007/s10745-014-9651-y>
- de Aranzabal I, Schmitz MF, Aguilera P, Pineda FD (2008) Modeling of landscape changes derived from the dynamics of socio-ecological systems. *Ecol Ind* 8(5):672–685. <https://doi.org/10.1016/j.ecolind.2007.11.003>
- Dogan S, Berktaş A, Singh VP (2012) Comparison of multi-monthly rainfall-based drought severity indices, with application to semi-arid Konya closed basin, Turkey. *J Hydrol* 470–471:255–268. <https://doi.org/10.1016/j.jhydrol.2012.09.003>
- Domptail S, Easdale MH, Yuerlita (2013) Managing Socio-Ecological Systems to Achieve Sustainability: A Study of Resilience and Robustness. *Env Pol Gov* 23(1):30–45. <https://doi.org/10.1002/eet.1604>
- Donatti CI, Nicholas K, Fedele G, Delforge D, Speybroeck N, Moraga P, Blatter J, Below R, Zvoleff A (2024) Global hotspots of climate-related disasters. *Int J Disaster Risk Reduct* 108:1–18. <https://doi.org/10.1016/j.ijdrr.2024.104488>
- Dutta P, Deka S, Debnath J, Sahariah D, Abouleish MY, Ali T, Chakraborty R, Singh SK, Kanga S, Kumar P, Meraj G (2026) SDG-relevant land-use change, transition intensity and event-based flood exposure assessment in the Lower Brahmaputra floodplain, India, 1990–2052. *Environ Dev Sustain*. <https://doi.org/10.1007/s10668-026-07763-x>
- Falloon P, Betts R (2010) Climate impacts on European agriculture and water management in the context of adaptation and mitigation—the importance of an integrated approach. *Sci Total Environ* 408(23):5667–5687. <https://doi.org/10.1016/j.scitotenv.2009.05.002>
- Folke C (2006) Resilience: The emergence of a perspective for social-ecological systems analyses. *Glob Environ Change* 16(3):253–267. <https://doi.org/10.1016/j.gloenvcha.2006.04.002>
- Gallopin GC (2006) Linkages between vulnerability, resilience, and adaptive capacity. *Glob Environ Change* 16(3):293–303. <https://doi.org/10.1016/j.gloenvcha.2006.02.004>
- Gedik F, Sarıç F (2026) Evaluation of groundwater drought in the Konya closed basin by examining hydrological and anthropogenic factors. *Arab J Geosci*. <https://doi.org/10.1007/s12517-025-12399-8>
- Gullacher, A., Allen, D.M., Goetz, J.D., (2023) Early Warning Indicators of Groundwater Drought in Mountainous Regions. *Water Resources Research* 59 (8) <https://doi.org/10.1029/2022WR033399>
- Hamed KH, Ramachandra Rao A (1998) A modified Mann-Kendall trend test for autocorrelated data. *J Hydrol* 204(1–4):182–196. [https://doi.org/10.1016/S0022-1694\(97\)00125-X](https://doi.org/10.1016/S0022-1694(97)00125-X)
- Hirsch RM, Alexander RB, Smith RA (1991) Selection of methods for the detection and estimation of trends in water quality. *Water Resour Res* 27(5):803–813. <https://doi.org/10.1029/91WR00259>
- Holling CS (1973) Resilience and Stability of Ecological Systems. *Annu Rev Ecol Syst* 4(1):1–23
- Hoogesteger J (2022) Regulating agricultural groundwater use in arid and semi-arid regions of the Global South: Challenges and socio-environmental impacts. *Curr Opin Environ Sci Health* 27:100341. <https://doi.org/10.1016/j.coesh.2022.100341>
- Hossain MS, Dearing JA, Rahman MM, Salehin M (2016) Recent changes in ecosystem services and human well-being in the Bangladesh coastal zone. *Reg Environ Change* 16(2):429–443. <https://doi.org/10.1007/s10113-014-0748-z>
- Humphrey V, Rodell M, Eicker A (2023) *Surv Geophys* 44(5):1489–1517. <https://doi.org/10.1007/s10712-022-09754-9>. Using Satellite-Based Terrestrial Water Storage Data: A Review
- Intergovernmental Panel on Climate Change (IPCC) (2023) Climate Change 2023: Synthesis Report. Contribution of Working Groups I, II and III to the Sixth Assessment Report of the Intergovernmental Panel on Climate Change [Core Writing Team, H. Lee and J. Romero (eds)]. IPCC, Geneva, Switzerland, p 184pp
- Kartal V, Nones M (2024) Assessment of meteorological, hydrological and groundwater drought in the Konya closed basin, Türkiye. *Environ Earth Sci*. <https://doi.org/10.1007/s12665-024-11587-1>
- Kendall MG (1975) *Rank Correlation Methods*, 4th edn. ed. Charles Griffin, London
- Koley S, Jeganathan C (2022) Evaluating the climatic and socio-economic influences on the agricultural drought vulnerability in Jharkhand. *Environ Monit Assess* 195(1):8. <https://doi.org/10.1007/s10661-022-10557-z>
- Lambin EF, Geist HJ, Lepers E (2003) Dynamics of Land-Use and Land-Cover Change in Tropical Regions. *Annu Rev Environ Resour* 28(1):205–241. <https://doi.org/10.1146/annurev.energy.28.050302.105459>
- Lankford B, Pringle C, McCosh J, Shabalala M, Hess T, Knox JW (2023) Irrigation area, efficiency and water storage mediate the drought resilience of irrigated agriculture in a semi-arid catchment. *Sci Total Environ* 859(Pt 2):1–14. <https://doi.org/10.1016/j.scitotenv.2022.160263>

- Li T, Dong Y, Liu Z (2020) A review of social-ecological system resilience: mechanism, assessment and management. *Sci Total Environ*. <https://doi.org/10.1016/j.scitotenv.2020.138113>
- Liu C, Yang C, Yang Q, Wang J (2021) Spatiotemporal drought analysis by the standardized precipitation index (SPI) and standardized precipitation evapotranspiration index (SPEI) in Sichuan Province, China. *Sci Rep* 11(1):1–24. <https://doi.org/10.1038/s41598-020-80527-3>
- Mann HB (1945) Nonparametric Tests Against Trend. *Econometrica* 13(3):245. <https://doi.org/10.2307/1907187>
- Mishra AK, Singh VP (2010) A review of drought concepts. *J Hydrol* 391(1–2):202–216. <https://doi.org/10.1016/j.jhydrol.2010.07.012>
- Murungweni C, van Wijk MT, Andersson JA, Smaling EMA, Giller KE (2011) Application Fuzzy Cogn Mapp Livelihood Vulnerability Anal E&S 16(4). <https://doi.org/10.5751/ES-04393-160408>
- Oliveira BM, Boumans R, Fath BD, Harari J (2022) Socio-ecological systems modelling of coastal urban area under a changing climate – Case study for Ubatuba, Brazil. *Ecol Model* 468:109953. <https://doi.org/10.1016/j.ecolmodel.2022.109953>
- Orduña Alegria ME, Zipper S, Shin HC, Deines JM, Hendricks NP, Allen JJ, Bohling GC, Golden B, Griggs BW, Lauer S, Lin C-Y, Marston LT, Sanderson MR, Smith SM, Whittemore DO, Wilson BB, Yu DJ, Yu QC, Butler JJ (2024) Unlocking aquifer sustainability through irrigator-driven groundwater conservation. *Nat Sustain* 7(12):1574–1583. <https://doi.org/10.1038/s41893-024-01437-0>
- Orhan O, Yakar M, Ekerin S (2020) An application on sinkhole susceptibility mapping by integrating remote sensing and geographic information systems. *Arab J Geosci*. <https://doi.org/10.1007/s12517-020-05841-6>
- Orhan O, Haghsheenas Haghighi M, Demir V, Gökkaya E, Gutiérrez F, Al-Halbouni D (2024) Spatial and temporal patterns of land subsidence and sinkhole occurrence in the Konya Endorheic Basin, Turkey. *Geosciences*. <https://doi.org/10.3390/geosciences14010005>
- Ortega-Gaucin D, La Cruz Bartolón J, Castellano Bahena H (2018) Drought Vulnerability Indices in Mexico. *Water* 10(11):1–32. <https://doi.org/10.3390/w10111671>
- Ostrom E (2009) A general framework for analyzing sustainability of social-ecological systems. *Sci (New York N Y)* 325(5939):419–422. <https://doi.org/10.1126/science.1172133>
- Pasho E, Camarero J, Vicente-Serrano J, S.M (2012) Climatic impacts and drought control of radial growth and seasonal wood formation in *Pinus halepensis*. *Trees* 26(6):1875–1886. <https://doi.org/10.1007/s00468-012-0756-x>
- Rani A, Gupta SK, Singh SK, Meraj G, Kumar P, Kanga S, Đurin B, Dogančić D (2023) Predicting Future Land Use Utilizing Economic and Land Surface Parameters with ANN and Markov Chain Models. *Earth* 4(3):728–751. <https://doi.org/10.3390/earth4030039>
- Ross A, Martinez-Santos P (2010) The challenge of groundwater governance: case studies from Spain and Australia. *Reg Environ Change* 10(4):299–310. <https://doi.org/10.1007/s10113-009-0086-8>
- Roy P, Pal SC, Chakraborty R, Chowdhuri I, Saha A, Shit M (2022) Climate change and groundwater overdraft impacts on agricultural drought in India: Vulnerability assessment, food security measures and policy recommendation. *Sci Total Environ* 849:1–18. <https://doi.org/10.1016/j.scitotenv.2022.157850>
- Roy D, Korir YC, Gillespie SA, Hossain MS (2025) Social-ecological systems approach in drought-food insecurity nexus research. *NPJ Sustain Agric*. <https://doi.org/10.1038/s44264-025-00070-4>
- Sanga U, Sidibé A, Olabisi LS (2021) Dynamic pathways of barriers and opportunities for food security and climate adaptation in Southern Mali. *World Dev* 148:105663. <https://doi.org/10.1016/j.worlddev.2021.105663>
- Santini M, Noce S, Antonelli M, Caporaso L (2022) Complex drought patterns robustly explain global yield loss for major crops. *Sci Rep* 12(1):5792. <https://doi.org/10.1038/s41598-022-09611-0>
- Sarı F, Kahveci M, Somay-Altas M, Tuşat E (2021) Evaluating sink-hole formation with multicriteria decision analysis: a case study in Karapınar-Konya, Turkey. *Arab J Geosci*. <https://doi.org/10.1007/s12517-021-06560-2>
- Scanlon BR, Longuevergne L, Long D (2012) Ground referencing GRACE satellite estimates of groundwater storage changes in the California Central Valley, USA. *Water Resour Res*. <https://doi.org/10.1029/2011WR011312>
- Serkendiz H, Tatlı H (2023) Assessment of multidimensional drought vulnerability using exposure, sensitivity, and adaptive capacity components. *Environ Monit Assess* 195(10):1154. <https://doi.org/10.1007/s10661-023-11711-x>
- Serkendiz H, Tatlı H, Özcan H, Çetin M, Sungur A (2023) Multidimensional assessment of agricultural drought vulnerability based on socioeconomic and biophysical indicators. *Int J Disaster Risk Reduct* 98:104121. <https://doi.org/10.1016/j.ijdrr.2023.104121>
- Serkendiz H, Tatlı H, Kılıç A, Çetin M, Sungur A (2024) Analysis of drought intensity, frequency and trends using the spei in Turkey. *Theor Appl Climatol* 155(4):2997–3012. <https://doi.org/10.1007/s00704-023-04772-y>
- Serkendiz H, Tatlı H, Özalkan E, Cetin M (2025) Machine learning and geographic information systems-based framework for multidimensional analysis of cascading drought impacts using remote sensing and in-situ data. *Sci Total Environ* 1001:180504. <https://doi.org/10.1016/j.scitotenv.2025.180504>
- Taşoğlu E, Öztürk MZ, Yazıcı Ö (2024) High Resolution Köppen-Geiger Climate Zones of Türkiye. *Intl J Climatology* 44(14):5248–5265. <https://doi.org/10.1002/joc.8635>
- van Vliet J, de Groot HL, Rietveld P, Verburg PH (2015) Manifestations and underlying drivers of agricultural land use change in Europe. *Landsc Urban Plann* 133:24–36. <https://doi.org/10.1016/j.landurbplan.2014.09.001>
- Vicente-Serrano SM, Beguería S, López-Moreno JI (2010) A Multiscalar Drought Index Sensitive to Global Warming: The Standardized Precipitation Evapotranspiration Index. *J Clim* 23(7):1696–1718. <https://doi.org/10.1175/2009JCLI2909.1>
- Vicente-Serrano SM, Gouveia C, Camarero JJ, Beguería S, Trigo R, López-Moreno JI, Azorín-Molina C, Pasho E, Lorenzo-Lacruz J, Revuelto J, Morán-Tejeda E, Sanchez-Lorenzo A (2013) Response of vegetation to drought time-scales across global land biomes. *Proc Natl Acad Sci USA* 110(1):52–57. <https://doi.org/10.1073/pnas.1207068110>
- Wilhite DA (2000) Chapter 1 Drought as a Natural Hazard: Concepts and Definitions. In: Wilhite DA (ed) *Drought: A global assessment*. Routledge hazards and disasters series. Routledge, London, pp 1–21
- Young OR, Berkhout F, Gallopin GC, Janssen MA, Ostrom E, van der Leeuw S (2006) The globalization of socio-ecological systems: An agenda for scientific research. *Glob Environ Change* 16(3):304–316. <https://doi.org/10.1016/j.gloenvcha.2006.03.004>
- Yu H, Lu N, Fu B, Zhang L, Wang M, Tian H (2022) Hotspots, co-occurrence, and shifts of compound and cascading extreme climate events in Eurasian drylands. *Environ Int* 169:1–15. <https://doi.org/10.1016/j.envint.2022.107509>
- Yue S, Wang CY (2002) Applicability of prewhitening to eliminate the influence of serial correlation on the Mann-Kendall test. *Water Resour Res*. <https://doi.org/10.1029/2001WR000861>

Zhu X, Wang G (2024) Impact of Agricultural Cooperatives on Farmers' Collective Action: A Study Based on the Socio-Ecological System Framework. *Agriculture* 14(1):96. <https://doi.org/10.3390/agriculture14010096>

Publisher's Note Springer Nature remains neutral with regard to jurisdictional claims in published maps and institutional affiliations.